FACTORS INFLUENCING CONSUMER TRUST IN AI CHATBOTS FOR ONLINE CUSTOMER SERVICE IN MALAYSIA

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DECLARATION

I hereby declare that this Research Project entitled:
"FACTORS INFLUENCING CONSUMER TRUST IN AI CHATBOTS FOR ONLINE CUSTOMER SERVICE IN MALAYSIA" is my own original work and has not been submitted previously or concurrently, in whole or in part, for any other degree or qualification at this or any other institution of higher learning.

All sources of information used in this research project have been duly acknowledged and referenced.

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ABSTRACT

The rapid adoption of artificial intelligence (AI) technologies has led to the increasing use of AI chatbots in online customer service. Despite their growing popularity, consumer trust remains a critical factor influencing the acceptance and continued use of AI chatbots. Therefore, this study aims to examine the factors influencing consumer trust in AI chatbots for online customer service in Malaysia. Specifically, the study investigates the effects of perceived usefulness, perceived ease of use, perceived interactivity, perceived risk, and anthropomorphism on consumer trust in AI chatbots.

This study employed a quantitative research approach using a structured questionnaire to collect data from consumers in Malaysia who have experience interacting with AI chatbots for online customer service. The collected data were analysed using IBM SPSS Statistics to evaluate the relationships between the independent variables and consumer trust.

The findings of this study provide insights into the key factors that contribute to building consumer trust in AI chatbots. The results are expected to assist businesses and service providers in improving the design and implementation of AI chatbot services to enhance customer trust and satisfaction. Furthermore, this study contributes to the existing literature on consumer trust and AI chatbot adoption within the Malaysian context.

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CHAPTER 1

INTRODUCTION

1.1 Overview

With the rapid advancement of AI chatbots, it has been widely used in many online services in Malaysia, especially in e-commerce sectors. This AI chatbots provide fast response with 24 hours availability, allows for product inquiries, order tracking, payment problem and after-sales-services, which improve the services efficiency and enhance customer satisfaction, making it an important communication tool between the seller and buyer. Although this AI chatbots provide convenience, but customers trust remains a critical factor influencing their effectiveness and acceptance. The degree to which consumers are ready to trust chatbots, largely depend on their degree of trust in the technology in an online service environment. In the e-commerce of Malaysia, users' trust in AI chatbot technology is not only driven by its technical aspects but to certain extent by a variety of subjective perceptions as well. This study aims to explore perceived usefulness, perceived ease of use, perceived interactivity, perceived risk, and anthropomorphism as the main predictors of consumer trust in the AI chatbots to serve customers online in Malaysia. This research is to contribute targeted perspectives to online retailers and system developers for the effective design of AI chatbots, consumer trust reinforcement and betterment overall online shopping experience through a focus on the Malaysian e-commerce sector.

1.2 Backgrounds of the study

The rapid advancement of Artificial Intelligence (AI) has led to the widespread deployment of AI-based chatbots in customer service, driven by their potential to enhance service efficiency and reduce operational costs (Prakash et al., 2023). In recent years, digital transformation is no longer a mere trend in business; it's considered a critical approach to enhance competitiveness, process and cost efficiency as well as customer satisfaction. Digital technologies, including artificial intelligence (AI), machine learning and data analytics, have increasingly helped organizations drive automation, decision making optimisation and delivery of personalised services. Among them, artificial intelligence has received the most attention, because of its ability to handle big data in a more effective manner and to mimic human decision-making processes, as well as its demonstrated robustness and efficiency in repetitive work. In the service field, AI has been widely used in various applications including intelligent recommendation systems in electronic commerce, and virtual assistant systems in finance, health care, telecommunications etc.

AI powered chatbots is one of the most symbolic and common forms of AI applications amid a variety of AI technologies. AI chatbots are increasingly adopted in online customer service to enhance efficiency and user experience (Ashfaq et al., 2020; Alagarsamy & Mehrolia, 2023). These chatbots streamline customer service interactions and deliver consumers with instant benefits, from answering frequently asked questions, to leading users through the process of services, and at times even making personalized recommendations based on past interactions profiles. AI chatbots will enable businesses provide a uniform level of quality service between different times and contexts by significantly lowering labor costs and improving responsiveness. Real life examples in different industries around the world have proved that AI chatbots contribute positively to boosting customer engagement, brand loyalty and overall satisfaction levels.

Customer service is one of the most important touchpoints between a company and its consumers, which has an impact on customer satisfaction, retention and loyalty (Gao et al., 2025). Legacy customer service systems are very human resources-heavy, requiring a great deal of time and money. They generally face problems of the services being slow or erratic at peak traffic but ok when idle. With the development of digital technology, AI chatbots had been gradually adopted either as a substitution to human agents or a supplement for process and service efficiency and reliability. They offer immediate answers to customer queries, can

resolve simple or frequent questions autonomously and, if needed, forward complex issues to human assistants in such a way that it results not only in staff-machine cooperation.

In e-commerce systems, where the customer interaction frequency is relatively high and there are large volume of enquiries and these days customer expect instant respond time from agent, such scenarios AI chatbots have most promising applicability. They assist consumers through various touchpoints like product discovery, order status inquiries, payment-related queries and returns processing and consistently collect and analyse the data about customer interaction. It helps companies to better understand the way in which consumers behave and prefer in purchasing. AI chatbots offer a level of service that is reliable, easy and somewhat personalised, while providing efficiency in online shopping and increased conversion rates.

While there is a lot of promise associated with them, whether AI chatbots will work in practice depends on how much consumers trust them. Trust is a cornerstone of human-AI interaction, especially in service contexts where consumers rely on chatbots for support (Li et al., 2023). Users must trust that a chatbot is being reliable with information and the system's reliability, data security, and problem-solving power. If trust declines, users interact less with the chatbot and thus have a lower overall satisfaction. This might cause them to stop using digital service channels entirely and seek help from their fellow humans.

This is a particularly prevalent phenomenon in Malaysia as its digital economy continues to grow; businesses here are employing such artificial intelligence (AI) chatbots in their customer service strategy, especially among those involved in e-commerce. In Malaysia, chatbots are gaining traction in e-commerce platforms such as Shopee and Lazada (Wong et al., 2023). The emergence of the COVID-19 pandemic powered up even more the use of AI-led customer service solutions. Faced with the twin pressures of increased demands for online transactions on one hand and constrained physical service delivery methods on another, firms felt that they had no option but to massively adopt automation technologies for them to continue serving and operating. Most Malaysian customers would prefer online shopping due to its convenience, product diversity and the price competitiveness, forcing the businesses to use AI chatbots for fast customer service (Gao et al., 2025). Although the AI chatbots can handle massive amount of data and provide efficiency, but challenges still occur. Most companies find this AI chatbots can save manpower and money while increasing uptime as it avoids dependent on human agents to respond to queries (Gao et al., 2025). They also gather information that can be used to customise shopping experiences and marketing campaigns. On the other hand, challenges

are user concerns about privacy, data security, mistakes in provision of services and ‘absence of human touch’ feeling.

Though AI chatbots are widely employed in various industries in Malaysia, the e-commerce sector is a setting of interest to study consumer trust since it involves online transactions, sharing personal information and interacting with customers regularly (Gupta et al., 2023). E-commerce is also a fierce competitive market where service quality plays an important role in the consumer’s decision to choose or stick with a particular e-tailer (Gupta et al., 2023). Although research into e-commerce chatbots is limited, with a particular interest in AI chatbots in Malaysian e-commerce, it is expected that this study will contribute specifically to the development of focused provisions in which businesses are able to improve on their chatbot design process in ensuring earning customer trust and transactions efficiency as well as company’s performance (Gao et al., 2025).

1.3 Research Problem

The concept of trust in technology-based systems has been extensively explored from a theoretical viewpoint like Technology Acceptance Model (TAM) and trust theory, focusing on the importance of perceived usefulness and ease of use for shaping acceptance and continued usage of technological applications (Herlina et al., 2025). Nonetheless, AI chatbots in e-commerce customer service gave rise to new interaction dynamics going beyond the previously known TAM constructs. Perceived interactivity, perceived risk and anthropomorphism have become more salient, as chatbots communicate automatically, give immediate responses and deal with important personal or transactional data. Although previous studies have often concentrated only on technology acceptance or user satisfaction (Herlina et al., 2025) and developing the understanding of how these factors combine to impact consumer trust in AI chatbots is scant, especially when it comes to e-commerce (Agarwal, 2025).

This study investigates consumer trust in AI chatbots used in Malaysia's e-commerce sector. In recent years, Malaysia's e-commerce industry has seen substantial growth, backed by three core driving factors. To respond to the surging demand for customer service, local e-commerce enterprises have widely deployed AI chatbots. These chatbots undertake four full-process core customer service functions doubtful (Fu et al., 2024). While they have two key core advantages, they still face two major sets of core concerns among consumers: some consumers question their ability to understand user needs, the practicality of the solutions they provide, and the smoothness of their service; another group of consumers cannot communicate smoothly due to hard-to-operate interfaces and responses that fail to meet expectations. Consumer trust is the core issue that restricts these technologies from delivering their intended practical effects and achieving long-term success (Mangipudi, 2025).

Most existing studies on consumer trust in AI chatbots take perceived usefulness and perceived ease of use as their core influencing variables (Fu et al., 2024). This research further expands the scope of impact factors to add three additional core variables: perceived interactivity, perceived risk, and anthropomorphism. Perceived interactivity refers to a chatbot's ability to deliver timely responses, conduct effective conversations, and maintain smooth communication. Perceived risk stems from users' concerns over privacy, data security, and misuse of their personal information. Anthropomorphism describes the human-like characteristics a chatbot displays through its conversational language, emotional expression,

and personalized interactions, traits that are more likely to boost user trust. These factors align with findings that AI assimilation plays a significant role in customer relationship quality, as AI can help build customer trust by providing faster and more accurate service and minimising errors when handling customer complaints (Hariguna & Ruangkanjanases, 2024). However, existing empirical studies have reached inconsistent conclusions about the impacts of the above variables (Fu et al., 2024). There is a lack of targeted research focused on Malaysia's e-commerce context. This study conducts its analysis specifically around this context. Theoretically, it supplements existing literature in the field of AI adoption and trust. Practically, it provides operational, actionable suggestions for local e-commerce enterprises to optimize their chatbot performance and strengthen consumer trust.

1.4 Research Question/s

The focus of this study is to investigate the associations between the independent variables that are chosen and consumer trust towards AI chatbots applied in online customer service associated with Malaysian e-commerce sector.

RQ1: What is the relationship between perceived usefulness and consumer trust in AI chatbots for online customer service in the Malaysian e-commerce sector?

RQ2: What is the relationship between perceived ease of use and consumer trust in AI chatbots for online customer service in the Malaysian e-commerce sector?

RQ3: What is the relationship between perceived interactivity and consumer trust in AI chatbots for online customer service in the Malaysian e-commerce sector?

RQ4: What is the relationship between perceived risk and consumer trust in AI chatbots for online customer service in the Malaysian e-commerce sector?

RQ5: What is the relationship between anthropomorphism and consumer trust in AI chatbots for online customer service in the Malaysian e-commerce sector?

1.5 Research Objective/s

This study's primary goal is to investigate the variables affecting consumers' trust in AI chatbots used for online customer support in Malaysia's e-commerce sector.

RO1: To investigate the relationship between perceived usefulness and consumer trust in AI chatbots for online customer service in the Malaysian e-commerce sector.

RO2: To examine the relationship between perceived ease of use and consumer trust in AI chatbots for online customer service in the Malaysian e-commerce sector.

RO3: To explore the relationship between perceived interactivity and consumer trust in AI chatbots for online customer service in the Malaysian e-commerce sector.

RO4: To study the relationship between perceived risk and consumer trust in AI chatbots for online customer service in the Malaysian e-commerce sector.

RO5: To analyse the relationship between anthropomorphism and consumer trust in AI chatbots for online customer service in the Malaysian e-commerce sector.

1.6 Justifications of the study

Theoretically, the present study also extends TAM in the context of AI chatbots for e-commerce by providing a better apprehension about technology acceptance and trust of consumer to technology. Whilst past research has predominantly focused on perceived usefulness and perceived ease of use to explain technology acceptance, such an approach may not be adequate in explaining trust building in the current AI-based service context. Accordingly, we extend the classic TAM model by supplementing theoretical factors such as perceived interactivity and perceived risk with the antecedent anthromorphism for understanding trust in AI supported customer service (Li et al., 2023). Empirically examining these independent variables with consumer trust, the current study fills a contextual void in extant literature and contributes with theoretical insights about how trust is established in automated, non-human service systems in the context of emerging digital economy like Malaysia.

From a practical and policy perspective, the findings of this research provide well-grounded implications for e-commerce companies, chatbot developers and policymakers practicing digital commerce but also consumer protection. In practice, the findings can support those e-

commerce platforms in understanding what are the most important factors to affect consumer trust and thus provide meaningful implications on how design and implement an AI chatbot more friendly, interactive, secure. This in turn can increase customer satisfaction, service efficiency and long-term customer loyalty (Gao et al., 2025; Gupta et al., 2023). From a policy standpoint, the findings will contribute empirically to assisting policymakers and regulatory bodies in developing policies guiding AI governance, data privacy, and consumer protection in e-commerce. Similarly, understanding consumer trust issues can help in formulating policies that support responsible AI deployment, promulgate transparency, and build consumer confidence in AI-driven online services (Gupta et al., 2023).

1.7 Research Scope

The present research is aimed at investigating the determinants of consumers trust on AI chatbots deployed in Malaysian e-commerce industry for online customer service. This study examines the links between perceived usefulness, perceived ease of use, perceived interactivity, perceived risk, anthropomorphism and consumer trust. The respondents for this study are Malaysian consumers who have used AI chatbots during shopping on e-commerce sites before. The study is carried out in Malaysia and focuses on users of diverse demographics who participate in online shopping. Collection of data is performed in a specific time that reflects current usage of AI chatbot and consumers' perceptions in the Malaysian e-commerce.

1.8 Definitions of the key terms

Key terms	Definition	Citation
Consumer Trust (DV)	Consumer trust on AI chatbot is that the consumer wants to trust an AI chatbot while talking in a conversation though not sure and believes it is risky based on how they perceive the skill, honesty, and reliability of this chatbot during interaction.	(Li et al., 2023)
Perceived Usefulness (IV1)	Perceived usefulness refers to the degree to which a person believes that using technology enhances performance of tasks or provides tangible benefits.	(Hoo et al., 2023)
Perceived Ease of Use (IV2)	Perceived ease of use is associated with perceived simplicity among consumers on how easy it is to learn and operate an AI chatbot and the level of physical and cognitive effort needed to reach certain goals.	(Li et al., 2023)
Perceived Interactivity (IV3)	Perceived interactivity refers to the extent to which users view their chatbot interaction as being responsive, communicative, and conductive for a meaningful two-way communication.	(Prakash et al., 2023)
Perceived Risk (IV4)	Perceived risk refers to how much the customers think that incurring a chatbot may cause them invasion and threat of their privacy, as well as any other adverse effects.	(Bouhia et al., 2022)
Anthropomorphism (IV5)	Anthropomorphism is the degree to which users ascribe human characteristics, behavior, emotions and or social attributes to non-human entities such as AI systems, robots, and chatbots.	(Chi & Hoang Vu, 2023)

Artificial Intelligence (AI)	AI is a technology that makes it possible for robots to emulate human cognition and behavior, so they can absorb knowledge, communicate with people and execute tasks on their own in digital environments like e-commerce websites.	(Ding & Najaf, 2024)
AI Chatbots	AI chatbots are smart virtual agents that can engage with the customer, share information or answers questions and solve a transaction to enhance online commerce experience and efficiency.	(Ding & Najaf, 2024)
Online Customer Service	Online customer service is using the internet to provide consumers with fast, convenient and personalized help in fulfilling their purchasing needs, answering product questions and solving any issues they experience through their interactions with a company.	(Hoo et al., 2023)
Technology Acceptance Model (TAM)	TAM provides the explanatory power between consumers' perceptions of usefulness and ease of use towards their attitudes, trust, adaption intention to AI-based customer service systems.	(Prakash et al., 2023)
Trust Theory	Trust Theory characterizes trust as the formation of belief-based expectations regarding an entity, which reflect one's intentions to depend on that entity when feeling insecure or under risk.	(Prakash et al., 2023)

1.9 Organisation of research

The structure of this study is based around a three chaptered formulation. Chapter 1 Introduction covers the research background, problem statement, research questions, purpose of the study, significance of the study and scope of the study and definition of terms. This chapter gives the comprehension about the research context and lays a basis for our study.

Literature review is presented in Chapter 2, where the theoretical underpinning and empirical evidence of AI chatbot and consumer trust in e-commerce are expounded. This chapter discusses applicable theories, explains pervious research of consumer trust and independent variables (perceived usefulness, ease of use, perceived interactivity, perceived risk, and anthropomorphism), through which the gap between technology acceptance model and.org theory will be discovered. The conceptual framework to the study is then formulated to end the chapter.

Moreover, Chapter 3 Research strategy describes the research methodology applied to fulfil the aims of the research. This chapter shows the research design, hypothesis generation, sampling scheme, data collection techniques, questionnaire development and analysis of data. It is in this chapter that the methodological soundness of the study is laid, and ground prepared for empirical analysis in later chapters.

Chapter 4 of this study will present the research findings derived from data collected via questionnaire surveys, which cover six statistical analysis areas including demographic analysis and behavioral analysis. This study uses the SPSS 30.0 statistical tool, with the core objective of clarifying the correlation between independent variables and consumers' trust in AI chatbots in the online customer service scenario.

Chapter 5 of this paper, titled "Conclusions and Recommendations", is positioned as the core concluding chapter of the study. This chapter first sorts out core findings and puts forward overall conclusions centered on the research's objectives and hypotheses. It covers knowledge implications, managerial implications, research limitations, and recommendations for future research, and summarizes the research contributions of this study, which focuses on the theme of consumers' trust in AI chatbots in the context of Malaysia's online customer service scenario.

CHAPTER 2

LITERATURE REVIEW

2.1 Overview

This chapter presents a brief review of related literature referring to trust in AI chatbots and online customer service in e-commerce. It starts with introducing the theories that underlie this research, then reviews existing studies from both theoretical and empirical point of view, related to consumer trust, perceived usefulness, perceived ease of use, perceived interactivity and perceived risk. The chapter then discusses prior work on AI chatbot adoption in e-commerce, focusing specifically on insights that relate to consumer behaviour and trust. Lastly, assertions made by previous empirical work that a lack of empirical studies exist on other facets associated with entrepreneurial leadership are presented to demonstrate the need for this study and justify the development of its conceptual framework and hypotheses.

2.2 Industry Perspectives

The global e-commerce business has experienced rapid expansion in the past few years, fuelled by rising internet penetration, mobile device use and technological advancements (Global E-Commerce Market, n.d.). E-commerce platforms are commonly adopting AI technologies including machine learning and natural language processing to automate tasks, tailor customer experiences and boost operational performance. A well-known AI implementation is that of the AI chatbots, they are more becoming part of customer service by generating instant replies, answering high number of queries, and diminishing dependency on classic human agents (Gao et al., 2025). Globally, there has been a proliferation of chatbots in business applications with many companies deploying these systems to drive greater customer involvement while simplifying post-sale assistance as part of a general digital and automation trends shaping how organisations interact with their customers (Vebrianti et al., 2025).

Although there are some promising benefits, the application of AI in e-commerce has its own difficulties. Global business intelligence shows that issues of data privacy, security and transparency are three major barriers to consumer adoption and trust (Global E-Commerce

Market, n.d.). For instance, a large share of consumers express concerns about what happens to their personal data in the context of AI systems again pointing to trust as one of the emergent main sticking points for more widespread course-grade adoption of AI solutions into customer-facing applications (Vebrianti et al., 2025). Further, they face technical challenges like difficult integration with obsolete systems, maintaining data quality, and addressing ethical issues related to algorithmic bias and fairness in AI outputs (Ghosh, 2025). These issues illustrate the importance of investigating consumer trust and perceived risk towards AI chatbots for continuing to development in e-commerce customer service (Ghosh, 2025).

This e-commerce growth is also taking place regionally with ASEAN becoming one of the fastest growing e-commerce markets, though certain levels of AI development are lacking (Global E-Commerce Market, n.d.). Most sellers are cognizant of the potential, but adoption is still low according to a new industry report across Southeast Asia on e-commerce sellers and AI implementation with Malaysia reporting lower AI integration compared to their counterparts in other parts of the region (Lazada Group & Kantar, 2025). In Malaysia, just a quarter of sellers have adopted AI in their operations due to factors such as cost factor, complexity of adoption and its limited internal expertise (Lazada Group & Kantar, 2025). That would indicate that challenges for AI implementations are still formidable at the company level (Ghosh, 2025), even as many (on an industry level) come to see how dockens and other AI technologies can make a difference.

From a domestic perspective, Malaysian e-commerce businesses are becoming more receptive to AI technologies such as chatbots as they roll out digital transformation initiatives (Global E-Commerce Market, n.d.). But the chasm between awareness and deployment of AI suggests there are ongoing operational and organisational challenges (Lazada Group & Kantar, 2025). For instance, these could involve cost considerations, the level of technology readiness and workforce preparedness to maintain and control AI devices that can indirectly impact how consumers trust and perceive AI tools. Furthermore, the service provider industry, where quality of continuing interaction, confidentiality and dependability are critical aspects in a service interaction, also amplifies the significance for understanding what drives consumer trust (Vebrianti et al., 2025). In combination these factors and challenges presented in the industry highlight the importance of both exploring perceived usefulness, ease of use, interactivity and risk (Gao et al., 2025) for furthering both academic knowledge and business practice in Malaysian e-commerce.

2.3 Theoretical Perspectives

The current study is based on two well-known theories, namely Technology Acceptance Model (TAM), and Trust Theory. These theories offer a robust conceptual basis for investigating consumer trust formation in AI chatbots deployed as online customer service in Malaysian e-commerce. TAM accounts for how consumers perceive and adopt new technology, while Trust Theory offers an explanation of how trust evolves from the system interaction that users have with systems characterized with uncertainty and risk.

2.3.1 Technology Acceptance Model (TAM)

The TAM was designed by Davis (1989) to describe user acceptance of information systems. TAM was derived from Theory of Reasoned Action (TRA), which explained individuals' behaviors by attitudes and intentions. Perceived usefulness and perceived ease of use were proposed by Davis as two fundamental determinants driving users' attitudes towards technology adoption (Dziak, 2024). In the years that followed, TAM was validated and extended in different technological settings such as e-commerce, mobile applications, AI-based systems (Herlina et al., 2025).

As digital technologies continue to evolve, so has the technology industry from basic information systems to complex AI-driven applications. At the beginning, systems nearly solely targeted a purely functional efficiency, meanwhile contemporary technologies like AI chatbots should emphasize interactivity-based personal support and autonomous decision-making. With the development of more sophisticated technologies that were also more consumer-oriented, traditional factors of acceptance shifted to original/psychological measurements and experiences especially in industries which were services-based, such as the e-commerce industry (Faiz et al., 2024).

TAM has been widely used in the context of user acceptance of IT, such as for online banking, e-commerce websites and mobile services and AI. In these applications, perceived usefulness corresponds to the degree to which a technology improves task completion, and perceived ease of use indicates how easy is it to operate. There have been many empirical studies that showed two constructs influence users' attitude, intentions, and actual use of technology mediated services significantly (Herlina et al., 2025).

In the case of AI based chatbots used for online customer service, TAM was of specific interest as consumers rate chatbot performance according to how efficiently and easy it is to use. PU and PEOU could directly influence online shoppers' trust belief on and continued to use of AI chatbot (Shahbandi, 2025). Building on the extended TAM, this paper explores how perception of usefulness and ease-of-use influence trust toward AI chatbot in e-commerce Malaysian platform (Shahbandi, 2025).

Despite its widespread use, TAM has been criticised for lack of explanatory power in a complex, AI-driven service context. The model places the emphasis predominantly on cognitive assessments and ignores emotional, social, and interpersonal aspects such as trust, risk perception, and human-like interface (Shahbandi, 2025). In an AI chatbot context, users might feel uncertainty and privacy anxiety, but also a certain need of social identity: the TAM model does not cover these aspects well (Shahbandi, 2025). Thus, it is justified to consist of the other constructs which are perceived interactivity, perceived risk and anthropomorphism in TAM for a better knowledge of consumer trust in AI chatbots (Faiz et al., 2024).

2.3.2 Trust Theory

Trust Theory is based on social psychology and organizational behaviour (Legood et al., 2023). Trust, among other things, is the willingness of one agent to become vulnerable to the actions of another agent with respect to some actions they take and about which there are positive expectations that independent decision taken by either participant could adversely affect them (Isaeva et al., 2020). Trust Theory has been also transposed to technology mediated spaces, and trust is not limited anymore to people but encompasses trust in system, platforms and AI likewise (Gulati et al., 2024).

As digital services matured, trust flourished around the web as the lack of face-to-face contact and new unknowns took hold (Isaeva et al., 2020). For e-commerce and AI services people have to be willing to believe the systems will work, keep their private data secure, and give them a good result. The development of AI chatbots has added another layer onto the issue of trust, as they independently interpret data, communicate with individuals and influence decision-making (Isaeva et al., 2020).

Trust Theory has been widely used in the investigation of user trust towards digital platforms including e-commerce, online banking, information systems and AI (Gulati et al., 2024). Trust in the above settings is often shaped by features like system dependability, quality of information, perceived risk and transparency. Trust heavily influences intention to use, satisfaction and customer loyalty (Isaeva et al., 2020).

In particular, Trust Theory is appropriate for this study as AI chatbots are functioning in environments which can be described by uncertainty, information asymmetry and privacy concerns (Isaeva et al., 2020). Consumers have to believe that the chatbots are offering them accurate information and not selling their personal data or violating their best interests. Perceived risk and anthropomorphism are in line with Trust Theory in the sense that they describe how vulnerability and uncertainty influence consumer reliance on AI chatbots in Malaysian e-commerce (Gulati et al., 2024).

Although Trust Theory offers a rich theoretical foundation for reasoning about vulnerability and dependence in the use of technology, it does not explicate how design factors impact trust-building in AI systems (Isaeva et al., 2020). Perception of trust towards AI chatbots could also be contingent upon interface design, quality of interaction and human-like cues that are not explicitly included in conventional Trust Theory (Gulati et al., 2024). As such, integrating Trust Theory and TAM enables this research to capture both utilitarian adoption and emotional trust-building processes in the context of AI chatbot use.

2.4 Consumer Trust in AI Chatbots

Consumer trust refers to consumer's readiness to entrust to an AI chatbot, through embracing vulnerability and uncertainty, informed by favorable cognitive appraisal of the chatbot's ability, benevolence, integrity and concern in service exchanges. (Prakash et al., 2023). Consumer trust in chatbots is users' confidence in an AI-driven chatbot for interaction with it to obtain services, based on-apparent AI reliability, perceived competency and human-like friendly behavior with the reduced level of ambiguousness during use (Alagarsamy & Mehrolia, 2023). Consumer trust is when a customer believes that an AI chatbot is trustworthy, it has credibility and it can stand for their best interest while dealing in online shopping transactions (Ding & Najaf, 2024).

Consumer trust is a key construct of interest in e-commerce and AI chatbot research as it directly affects user engagement, adoption and retention (Gao et al., 2025; Gupta et al., 2023). And high confidence would naturally translate to reduced uncertainty and perceived risk, resulting in satisfaction and continued use of this AI service (Isaeva et al., 2020; Li et al., 2023). If users don't trust, they are very likely to leave there will be a fallback on human support and direct sales transactions on the e-commerce can be hit (Gao et al., 2025).

Trust was developed in line with research from social psychology and organisational behaviour, concentrating on interpersonal and institutional links(Legood et al., 2023). Along with internet transactions and technological services running, the study extended to e-commerce and technological trust including on reliability, security and privacy (Gulati et al., 2024; Isaeva et al., 2020). Recently, trust in AI chatbots has been identified as a new topic, merging cognitive evaluative and emotional relationships especially from the perspective of virtual interactivity without human interaction (Alagarsamy & Mehrolia, 2023; Li et al., 2023).

Although a lack of consumer trust can be a barrier to new technology, the measurement and understanding of trust in AI systems is difficult (Gulati et al., 2024). Moreover, as AI chatbots lack human-like attributes like empathy, contextual comprehension and adaptive judgment that may influence cognitive trust perception in unpredictable ways (Chi & Hoang Vu, 2023; Shahbandi, 2025). Paradoxically, trust is also contextual. What instils trust with one online platform or service may not apply in another e-commerce scenario or culture (Fu et al., 2024; Isaeva et al., 2020). Besides, given the significance of trust had been demonstrated in earlier research, minimal empirical work had described how cognitive judgments and affective feelings combined to form it particularly in Malaysia's e-commerce context (Gupta et al., 2023).

2.5 Independent Variables

2.5.1 Perceived Usefulness

Perceived usefulness is the user's perception that a technology will provide functional benefit or enhance task performance (Prakash et al., 2023). Perceived usefulness is a user's belief that the technology will be beneficial in their activities (Alagarsamy & Mehrolia, 2023). Perceived usefulness is defined as users' belief that a system is useful to their tasks and increases the task performance (Rafiq et al., 2022).

Perceived usefulness is particularly crucial in the e-commerce sector, when consumers make a purchase decision or have any post-purchase issues, they expect a quick and efficient service in the era of e-commerce (Gao et al., 2025). AI chatbots can make online shopping experience smoother as they are perceived to be useful that provide quick answers, correct information and solve the issue (Prakash et al., 2023). On the other hand, customers become frustrated and lose faith in automated customer support systems when chatbots don't live up to their expectations or give incorrect answers (Fu et al., 2024).

This construct was initially proposed in the Technology Acceptance Model (TAM) to predict employees' acceptance of information systems and has since been adapted to different digital demands for consumers (Herlina et al., 2025; Faiz et al., 2024). In the era of artificial intelligence, perceived usefulness in the context of chatbot is primarily focused on a system's ability to comprehend user needs and provide efficient task-oriented help (Shahbandi, 2025).

There is empirical evidence indicating a positive correlation between perceived usefulness and consumer trust towards AI-based artefacts. It has been asserted in the literature that those systems that improve job performance, save time and deliver custom-made and flexible services are considered as being more useful that leads to higher user evaluations and greater trust (Alagarsamy & Mehrolia, 2023; Li et al., 2023). Studies in e-commerce and m-commerce also show that perceived usefulness is an important antecedent of users' trust toward online services (Gupta et al., 2023; Herlina et al., 2025).

However, perceived usefulness has limitations when used in AI chatbot studies. The construct predominantly emphasizes in-terms of functional and task-oriented performance, with limited attention is given to affective and social factors that must also contribute to cultivating trust (Chi & Hoang Vu, 2023; Shahbandi, 2025). In addition, usefulness perceptions are highly

dependent on the task at hand, and users can perceive a chatbot as being useful in different ways depending on the nature and complexity of the service that is underway (Faiz et al., 2024).

2.5.2 Perceived Ease of Use

Perceived ease of use is user perception that a technology can be used easily, without effort (Hoo et al., 2023). Perceived ease of use reflects users' perception that a system is simple, intuitive, and easy to operate (Alagarsamy & Mehrolia, 2023). Perceived ease of use is the extent to which customers believe it is easy for them to interact with a customer service chatbot without any hassle and performing service-related tasks (Prakash et al., 2023).

Perceived ease has an important role to play in online environments especially in e-commerce, where customers prefer fast and easy experiences (Gao et al., 2025). When these chatbots are in the form which is user friendly, give responses and be easy to use consumer tends to make use of it (Fu et al., 2024). In contrast, complicated or misunderstood interactions can cause frustration and reduce user engagement in the service (Alagarsamy & Mehrolia, 2023; Shahbandi, 2025).

It is derived from the TAM, which was originally developed to explain a person's intention to use new technology in a work context. It has subsequently been expanded to e-commerce sites, mobile apps and artificial intelligence systems (Herlina et al., 2025; Faiz et al., 2024). With chatbots, it really comes out during natural conversation, helping guide the user and make everything easy and stress free (Shahbandi, 2025).

Empirical evidence from prior research also substantially supports a positive relationship between perceived ease of use and trust towards digital or AI-based services. Previous studies show that easier-to-use systems can lower users' uncertainty and perceived effort, to the extent of encouraging more positive attitudes and a higher trust in them (Li et al., 2023; Prakash et al., 2023). Especially in AI chatbots, usability has been found to have significant effects on users' initial trust, satisfaction, and continuance intention (Fu et al., 2024).

It has some shortcomings in isolation. Just because a chatbot is accessible, user-friendly, or convenient doesn't mean resolution of problems or emotions will necessarily be effective (Chi & Hoang Vu, 2023). As the level of users' experience in digital technologies increase, ease of use may have less impact relative to other factors like perceived usefulness, interactivity and perceived risk (Herlina et al., 2025; Faiz et al., 2024). Thus, when considering the e-commerce climate of Malaysia, perceived ease of use could be considered in conjunction with other factors to have a clearer perspective on what influences consumer trust in AI chatbots.

2.5.3 Perceived Interactivity

Perceived interactivity refers to the extent which consumers believe a technology is responsive and reciprocal in its communication (Rafiq et al., 2022). Perceived interactivity refers to user perceptions of the responsiveness and dialogic nature of a technology as users interact (Ding & Najaf, 2024). Perceived interactivity is the general sense that users have of how responsive and involving a technology feels to use (Prakash et al., 2023).

In AI chatbot customer service, perceived interactivity is considered very important because it affects how engaged to and trusting of AI chatbot users are (Li et al., 2023). Compared to conventional interfaces, chatbots can engage users through natural and contextually relevant interactions that resembles human-like conversation (Alagarsamy & Mehrolia, 2023). High interactivity increases the user's sense of control and decreases frustration and increase satisfaction in use and by improving service efficiency and experience (Gao et al., 2025; Prakash et al., 2023).

The idea of interactivity came from communication and human–computer interaction theories so the concept of interactivity has been used to signify the relationship between users and media but remains a definitive quality in user experience design in e-commerce business process (Gupta et al., 2023). With the emergence of AI, interactivity in chatbots lies basically in understanding users' intent, providing intelligent responses and keeping conversational fluency (Ding & Najaf, 2024; Shahbandi, 2025).

Evidence from practice shows a consistent positive correlation between perceived interactivity and consumer trust, particularly in digital and AI services. Previous research suggests that interactive systems increase trust regarding the extent to which they are perceived as transparent, responsive, and reliable (Alagarsamy & Mehrolia, 2023; Li et al., 2023). In the realm of AI chatbots, more perceived interactivity has led to a higher trust level, satisfaction and intention to reuse chatbot for customer service (Fu et al., 2024).

It also has limitations. A high degree of interactivity might cause users to hold exaggerated accuracy and emotional understanding expectations that bot cannot always meet, leading to disappointment and reduced trust (Chi & Hoang Vu, 2023; Shahbandi, 2025). At the same time, an excessive level of interaction can infuriate someone looking for a quick answer (Gao et al., 2025). Thus, in Malaysia's e-commerce context the trade-off between interactivity and efficiency of precision is crucial for building consumer confidence.

2.5.4 Perceived Risk

The extent to which consumers determine that using an AI-based chatbot might cause unwanted or unknown consequences is called perceived risk (Li et al., 2023). Perceived risk is the subjective assessment of damages that can be caused while using banking chatbots (Alagarsamy & Mehrolia, 2023). Perceived risk is the user's level of belief that he or she will not face adverse consequences due to privacy and security aspects while using a chatbot (Prakash et al., 2023).

Perceived risk is an important consideration for consumer trust in chatbots, especially where personal and financial details are exchanged such as via e-commerce (Fu et al., 2024; Gupta et al., 2023). Concerns over privacy invasion, misinformation or data abuse may discourage users from using bots or trusting them to automated decision-making (Gao et al., 2025).

The concept of perceived risk comes from the consumer behaviour literature, where it was applied to explain uncertainty and anxiety regarding consumer purchase decisions, but was later transferred over to digital services like online shopping or e-banking as well and in that context also included issues with security, privacy and performance (Gupta et al., 2023; Isaeva et al., 2020). Considering the development of AI, in chatbot research perceived risk mainly includes data security, algorithm error, transparency and ethics issues (Chi & Hoang Vu, 2023; Gulati et al., 2024).

Consistently, empirical evidence shows a negative correlation between perceived risk and the trust of consumers in digital and AI-based services. Existing studies have shown that perceived risk has a negative effect on trust, satisfaction and usage intention toward chatbots (Alagarsamy & Mehrolia, 2023; Fu et al., 2024). Privacy and security concerns can, for example be reasons for trust violation with chatbot system in online banking and e-commerce scenario (Gupta et al., 2023; Li et al., 2023).

Risk perception is important, but its importance varies much across individuals, being also shaped by literacy, experience and cultural background aspects as well as characteristics of the risk itself (Isaeva et al., 2020). Particularly where consumers perceive that there are strong functional benefits related to chatbot use (Faiz et al., 2024), convenience and efficiency may take precedence over security concerns. Therefore, in the context of Malaysian e-commerce, consumer trust in chatbot use during an online sales encounter is perceived to be a second-order construct that consists of perceived risk and utility, ease of use, and interactivity factors (Gao et al., 2025; Gupta et al., 2023).

2.5.5 Anthropomorphism

Anthropomorphism refers to the extent to which an AI chatbot is perceived by users as possessing human-like attributes, behaviours, emotions, or social cues during interaction(Li et al., 2023). Anthropomorphism can be described as the belief that an AI chatbot displays humanlike characteristics, which provide them with the ability to engage in social interaction and emotional exchange with a user (Rafiq et al., 2022). The anthropomorphism is an indicator showing how much users think that AI devices have humanlike characteristics while they are communicating (Chi & Hoang Vu, 2023).

Personification is the key to influencing users' psychological reactions towards AI chatbots. Human-like chatbots can bring a better social presence because of their human features, which means that the interaction appears more natural and less alienating for the user who have difficulty accepting bots as non-human (Alagarsamy & Mehrolia, 2023; Li et al., 2023). Without being able to establish face-to-face contact with the user, anthropomorphic design decreases possible uncertainty in the communication, enhances effective communicative experiences and supports users trust and rely on a system (Chi & Hoang Vu, 2023; Prakash et al., 2023).

Anthropomorphism comes from social psychology and human computer interaction research, which describes that people are inclined to attribute human properties in the interface of non-human objects (Legood et al., 2023). With development of AI and NLP, chatbots have been able to create more human-like speech and emotions (Ding & Najaf, 2024). In the world of e-commerce, developers also improve interaction experiences by providing chatbots with anthropomorphic names, avatars and more empathic communication styles (Gao et al., 2025; Gupta et al., 2023).

The literature has found empirical evidence for a positive relationship between anthropomorphism and consumer trust of AI chatbots. Human-like chatbots are typically perceived by users as more trustworthy and to have a social presence (Li et al., 2023; Rafiq et al., 2022). Anthropomorphic relational behaviour and empathetic responses lessen ambiguity, increase emotional involvement leading to trust, satisfaction, use continuance (Alagarsamy & Mehrolia, 2023; Fu et al., 2024).

Anthropomorphic is also limited in some aspect of its use. Overuse or misuse of anthropomorphism can potentially place users at an above realistic expectation level on what a chatbot is capable of when the system doesn’t deliver, however could lead to disappointment

and betrayed trust in return (Chi & Hoang Vu, 2023; Shahbandi, 2025). At the same time, users from different cultures are willing to accept anthropomorphised AI, and ethical issues need to be considered (Gulati et al., 2024).

2.6 Summary

This chapter has examined the background literature on consumer trust in AI chatbots, specifically to identify the factors that influence trust of individuals in online customer service environments. In view of existing literature and recent evidenced-based research, the literature review identified five important influential factors that determine trust in AI chatbots: perceived usefulness (PU), perceived ease of use (PEOU), perceived interactivity (PI), perceived risk (PR), and anthropomorphism. These constructs were chosen because they have strong theoretical and empirical support from previous studies in technology acceptance, human–computer interaction and AI-based service systems.

Literature review has established the basis upon which the research framework can be developed by enlightening definitions, roles and relationships among the determined independent and dependent variables (consumer trust). Prior research has established that perceived usefulness, ease of use, interactivity and anthropomorphism have a positive effect on consumer trust while perceived risk is negatively correlated with the latter. By synthesizing these findings, the literature supports the development of a comprehensive conceptual framework that elucidates how both functional and social dimensions of AI chatbots influence consumer trust in e-commerce settings.

In addition, the review points to areas where more work is needed such as fusion of anthropomorphism and trust in chatbot-based customer service in a Malaysian context. Hence, the current research model includes these five variables to offer a comprehensive understanding on consumer trust towards AI chatbot in providing online customer service in Malaysia. This framework is then used in the development of hypotheses so that this theory-informed model structure becomes integrated with the research methodology adopted here.

CHAPTER 3

RESEARCH METHODOLOGY

3.1 Overview

This chapter describes the methodology employed to investigate the relationships set forth in Chapter 2. Drawing on the conceptual framework and research hypothesis derived from the literature, chapter 3 describes the research design, data collection methodology, sampling procedure perception statements instruments of measurements as well data analysis techniques used in the study. The methodology is employed to gauge the effect of perceived usefulness, perceived ease of use, perceived interactivity, perceived risk and anthropomorphism on consumer trust towards AI chatbots for online customer service in Malaysia. By applying well-tested quantitative methods, it makes sure the results are reliable and valid and gives a strong methodological base to test the research framework.

3.2 Research Framework

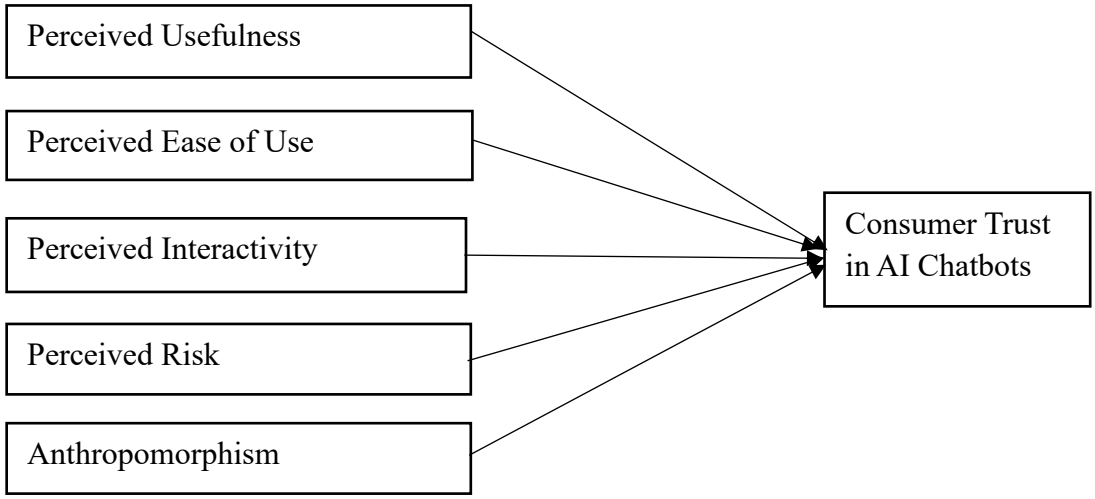
The perspective in which this research is framed was derived from the combination of TAM and Trust Theory as specified in Section 2.3. The two theories lay the theoretical ground for the understanding how trust is created in AI chatbot by the consumer context of Malaysian e-commerce online customer service. TAM accounts for users' technology acceptance by focusing on cognitive perceptions of usefulness and ease of use (Herlina et al., 2025), while Trust Theory predicts users' trust in a system under conditions of uncertainty and perceived risk (Isaeva et al., 2020). By combining these views, the theoretical framework encompasses functional and psychological antecedents of AI chatbot consumer trust.

Viewed from the theoretical aspect of TAM, perceived usefulness and perceived ease of use are two central concepts impacting one's perceiving on AI chatbots effectiveness and usability. According to e-commerce, consumers are likely to trust in the chatbot that is supposed to improve efficiency of service process; providing correct information and change problem-solving task easier (Prakash et al., 2023). These two constructs are instrumental and usability

assessment which underlies cognitive elements of trust towards AI-based customer service technologies.

Unlike the original TAM, this research goes beyond by including perceived interactivity, perceived risk and anthropomorphism which are also particularly pertinent in AI chatbot contexts. Perceived interactivity is the quality of two-way communication between a user and a chatbot, affecting users' engagement, sense of control and confidence regarding the system (Rafiq et al., 2022). Perceived risk, rooted in Trust Theory, embodies users' privacy or security concern and data abuse and failure of service concerning feeling that might decrease user acceptance to use AI chatbot (Li et al., 2023). Anthropomorphism is a term used to refer the degree that chatbots exhibit human-like qualities like social presence and empathetic responses that lower uncertainty and create emotional trust in no-human agents(Chi & Hoang Vu, 2023; Rafiq et al., 2022).

Figure 3. 1: Research Framework



Composing all these five IVs together i.e., perceived usefulness, perceived ease of use, perceived interactivity, perceived risk and anthropomorphism may be considered as affecting factors which are supposed to impact on trust in AI chatbots by consumers, taken as DV in the present study. Consumer trust represents users' belief in how qualified the chatbot, how reliable and how surer that it will act on their behalf through online service transaction (Ding & Najaf, 2024). The model proposes that trust in AI lies on the positive results of such constructs as perceived usefulness, ease of use, interactivity and anthropomorphism, whereas risk would negatively reduce consumer trust.

Finally, the proposed model offers an organized articulation of the connections among the independent variables and consumer trust in AI chatbots for e-retail customer service. Based on TAM and Trust Theory, the framework provides a theoretical model for guiding this study, and the foundation for formulating research hypotheses in next section. Theoretical concepts and empirical exploration are integrated through the model which guarantees consistency among theory, study design and hypothesis testing.

3.3 Hypotheses Development

A hypothesis is an empirical, testable proposition that proposes one or more correlation between two or more variables of the population being studied on the basis of existing knowledge previously determined from reading literature and forming our theoretic concept (Hypothesis-PhD Coursework, n.d.). In the statistical or quantitative research, hypotheses are basis for testing and determining if the relations in which they claim as relationship of two things are true empirically (Hypothesis-PhD Coursework, n.d.). This study takes conjecture approach to investigate the factors affecting consumers' trust in AI chatbots used for online customer support in Malaysia e-commerce industry.

Drawing on the conceptual model from Section 3.2, as well as incorporating TAM and Trust Theory, a total of five hypotheses are developed in this study. Each of the hypotheses is testing a relationship between one independent and dependent variable. Independent variable such as perceived usefulness, perceived ease of use, perceived interactivity, perceived risk and anthropomorphism have been of the hypothesis. These hypotheses are formulated by drawing on the theory and empirical evidence of CHAPTER 2.

The consistent dependent variable to which the hypotheses are all tied is consumer trust in AI chatbots. Consumer trust is the consumer's confidence in using AI chatbots for online customer service according to the perceived reliability, expertise, honesty and benevolence of the chatbot (Ding & Najaf, 2024). In e-commerce contexts where transactions are risky and uncertain, trust is a key factor influencing users' acceptance, usage, and continued use of AI chatbot services (Alagarsamy & Mehrolia, 2023; Li et al., 2023).

3.3.1 Perceived Usefulness and Consumer Trust in AI Chatbots

Perceived usefulness is the degree to which consumers believe that using an AI chatbot adds to their service experience by saving time and effort, increasing accuracy and improving task performance (Prakash et al., 2023). In e-commerce domain, timely and relevant responses as well as effectiveness in problem resolution can lead to high perceived usefulness of a chatbot (Gao et al., 2025).

Some studies in the past have shown that perceived usefulness is significantly related to the consumer trust on technology-based service. Studies in AI- and e-commerce-context confirm this, showing that systems are more likely to be trusted if they are seen as useful (Rafiq et al., 2022; Prakash et al., 2023) because of the associated prospect of tangible benefits and reduction in uncertainty during service interactions. When AI chatbots do an efficient job of assisting the consumers across their online purchase journey, users are more likely to gain confidence in the system for being able to guide them effectively.

From a critique point of view, whilst perceived usefulness represents the instrumental value in AI chatbots promising valuable and reliable services, it may not cover all emotional or social aspectual aspects of trust (Chi & Hoang Vu, 2023). Nevertheless, in the context of e-commerce settings that require speed and accuracy perceived usefulness continues to play an important role in influencing consumers' trust (Herlina et al., 2025). Consequently, likely to simultaneously affect perceived usefulness and consumer trust in AI chatbots.

H1: There is a relationship between perceived usefulness and consumer trust in AI chatbots for online customer service in Malaysia.

3.3.2 Perceived Ease of Use and Consumer Trust in AI Chatbots

Perceived ease of use is the degree to which consumers believe using an AI chatbot is free of effort and straightforward (Hoo et al., 2023). In online customer support systems, the perceived ease of use and the minimal effort required by users are critical to influence favourable user experiences in chatbot interaction (Prakash et al., 2023).

Previous empirical research indicates that perceived ease of use plays an important role in consumer trust toward digital and AI-enabled services. The lower level of uncertainty results in expanded trust boundaries the easier to interact between users is (Alagarsamy & Mehrolia, 2023; Hoo et al., 2023). In the context of chatbot interactions, smoothness also reduces cognitive load and frustration, which contributes to the increased acceptance and trust in the system (Fu et al., 2024; Shahbandi, 2025).

It is important to note that perceived ease of use alone may not lead to trust when the chatbot does not provide accurate or useful responses (Chi & Hoang Vu, 2023). Nevertheless, such convenience of use is becoming an increasingly important factor in the development of primary trust towards AI chatbots within a Malaysian e-commerce environment where different platforms and user interfaces are pulled together (Gao et al., 2025; Herlina et al., 2025).

H2: There is a relationship between perceived ease of use and consumer trust in AI chatbots for online customer service in Malaysia.

3.3.3 Perceived Interactivity and Consumer Trust in AI Chatbots

The level of perceived interactivity indicates the extent to which consumers feel that AI chatbots are accommodating, interactive, and can engage in dialogues (Prakash et al., 2023; Rafiq et al., 2022). In customer service contexts, engagement and control are further increased when interactive chatbots can determine the user intent, answer relevant information and maintain conversational flow (Ding & Najaf, 2024).

The extant literature shows that perceived interactivity is positively associated with consumer trust in digital service settings. Research has found that the use of interactive systems is also interpreted as more transparent, reliable, and human-like which promotes trust (Li et al., 2023; Prakash et al., 2023). For applications of AI chatbot service, more interactivity helps to reduce ambiguity and increase the level of confidence users have towards their screen-based dialogue systems (Alagarsamy & Mehrolia, 2023; Fu et al., 2024).

In terms of criticism, if the chatbot is unable to replicate human comprehension sufficiently, too much interactivity may result in unfulfilled expectations (Chi & Hoang Vu, 2023; Shahbandi, 2025). However, a certain degree of interactivity is required in establishing trust, particularly in e-commerce settings where quality of service depends heavily on communication effectiveness (Gao et al., 2025; Gupta et al., 2023).

H3: There is a relationship between perceived interactivity and consumer trust in AI chatbots for online customer service in Malaysia.

3.3.4 Perceived Risk and Consumer Trust in AI Chatbots

Perceived risk is a consumer's feelings about the possible bad results that they might get from using AI chatbots, such as privacy invasion and data misuse, misinformation or service failure (Bouhia et al., 2022; Li et al., 2023). Perceived risk is likely to be a concern for e-commerce transactions because of the use of personal and financial data (Gupta et al., 2023; Isaeva et al., 2020).

There is no consistent picture in empirical studies, however, on the interplay of perceived risk and consumer trust in AI-based services. The greater the perception of risk, users are more anxious and more reluctant to trust in automatic systems (Bouhia et al., 2022; Li et al., 2023). In chatbot conversations, issues of security and transparency can pose great threats to trust (Gulati et al., 2024; Li et al., 2023).

Most importantly, the influence of perceived risk may depend on users' experience and technological expertise (Isaeva et al., 2020). Unfortunately, in the environment of Malaysian e-commerce as a matter of privacy increasingly being brought to the attention of consumers, perceived risk is still a big problem when it comes to trust (Fu et al., 2024; Gupta et al., 2023).

H4: There is a relationship between perceived risk and consumer trust in AI chatbots for online customer service in Malaysia.

3.3.5 Anthropomorphism and Consumer Trust in AI Chatbots

Anthropomorphism is the degree of humanlike qualities consumers attribute to AI chatbots like social presence or empathy and conversational naturalness (Chi & Hoang Vu, 2023; Rafiq et al., 2022). Human-like characteristics could make chatbot chats feel less robotic and more personal (Ding & Najaf, 2024; Prakash et al., 2023).

Prior research suggests that anthropomorphic cues in interfaces may increase trust toward AI chatbots by enhancing psychological closeness and decreasing perceived uncertainty during human–AI interactions (Chi & Hoang Vu, 2023; Rafiq et al., 2022). Empathetic responses, natural language patterns and human conversational style of chatbot are likely to inspire emotional involvement as a potential influence to an enhanced trust towards the system (Li et al., 2023; Fu et al., 2024).

From a critical standpoint, anthropomorphism may increase trust; however, a too much human-like appearance might provoke over idealization and ethical issues (Gao et al., 2025; Shahbandi, 2025). However, when used correctly, anthropomorphism has clear potential as a trust- building technique for nonhuman service agents (Gupta et al., 2023; Prakash et al., 2023).

H5: There is a relationship between anthropomorphism and consumer trust in AI chatbots for online customer service in Malaysia.

3.4 Research Design

The design of the study is the strategy that the researcher intends to pursue, that shows how he will carry out the research to obtain an answer for his research questions and test his hypotheses methodically (Combes, 2021). It also describes the methods for data collection, measurement and analysis to ensure that the research questions of this study will be answered valid, therefore reliability. The research design is detailed, and it increases the trustworthiness of findings by maintaining a fit among the research question, theory, and method (Williams, 2024).

Broadly, research designs fall into qualitative, quantitative and mixed methods. The difference between quantitative research designs which are interested in measurement of variables on numeric scales, and testing relationships using statistics, and qualitative research designs that focused on deep understanding of meanings and experiences becomes evident in this context. Mixed methodology incorporates both approaches to obtain a fuller picture of the research phenomenon (Ahmed & Pereira, 2024).

This research has adopted a quantitative design to scrutinize the association of perceived usefulness, perceived ease of use, interactivity, bias risk defence mechanism and anthropomorphism with consumer trust for AI chatbots for customer service via online in Malaysia. A quantitative methodology is suitable for this research as the aims are to verify hypothesised relationships between unambiguously defined variables and determine whether these are statistically significant. The benefit with quantitative is if there's generalizable results, can be used everywhere, and empirical testing of the theoretical model would be nice to have (Ghanad, 2023).

This study utilizes a cross-sectional survey research design as data are collected once from respondents. Cross-sectional designs are often utilized in e-commerce and technology adoption research, as such design allows researchers to gauge consumers' immediate perceptions and attitudes about emerging technologies like AI chatbots(Gao et al., 2025). This methodology is appropriate for testing of consumer trust, because perceived trust can be easily influenced by new technological advances and user experiences.

A survey-based approach was utilized as it enables data to be rapidly collected from many Malaysian consumers, who have previous experience with AI chatbots in e-commerce. Standardized measurement of latent constructs and survey validation possible to measure a construct such as trust, usefulness and perception of risk with consistent Likert-scale items ensuring that responses are comparable among the respondents(Pratama & Harsono, 2025).

Moreover, survey techniques are relatively inexpensive and feasible for investigating populations that are in distant regions.

In addition, the quantitative cross-sectional method is congruent with the theoretical basis of this research. The Technology Acceptance Model (TAM) and Trust Theory have been widely empirically validated through quantitative survey methods in previous research, particularly studies on technology acceptance and trust of digital services (Prakash et al., 2023). Use of the same design increases methodological rigour and comparability of results with previous studies.

In sum, the use of a quantitative cross-sectional survey design to successfully address the research objectives and hypotheses is recommended for this study. The above approach allows systematic measurement and statistical analysis of independent variables effects on consumer trust, thus establishing a stable base to review how trust is formed in the context of AI chatbots in Malaysia e-commerce setting.

3.5 Sampling Design

Sampling Design

A procedure for choosing a sample of elements from a population and how to use the sample for purposes of making conclusions about the population. The correct sampling design guarantees that the sample chosen represents well the properties of the population, thereby increasing the generalizability and validity of study findings (Makwana et al., 2023). For quantitative studies, a well-designed sampling strategy is fundamental to reduce sample bias and facilitate meaningful statistical assessment.

Target Population

The focal population group for the study is Malaysian consumers who have experience interacting with AI chatbots via online customer service in e-commerce applications. This population is suitable as we are interested to explore consumer trust in AI chatbots in the context of online shopping in the Malaysian setting. AI chatbot users are perceived to be potential raters of AI chatbot usefulness, ease of use, interactivity, anthropomorphism outstanding as well as perceived risks. A well-formed target population will help data to be relevant to the research objectives (Makwana et al., 2023).

Unit of Analysis

The consumer is the focus of analysis in this investigation. Each interviewee is a unit of analysis and describes his/her personal views, impressions and experiences on the use of AI chatbots for internet customer service. The choice of the individual as the unit of analysis is in line with previous technology adoption and consumer behaviour research where attitude, perception and trust have been observed at an individual level (Innerview, 2024).

Sampling Frame

A frame of reference is the list or writer from whom a sample is taken (McCombes, 2023). However, all Malaysian consumers who have utilised AI chatbots in e-commerce are incomplete and exhaustive sampling frame was not easily accessible. Hence, the sampling frame can be from online platforms such as social media networks, online shopping communities; and even university student and working adult network possessing potential respondents which meet the study's criteria. This practice is popular among e-commerce research and artificial intelligence related studies due to the fact that their target population is digital.

Sample Size

A proper sample size calculation is essential for achieving adequate power and reliable results. According to (Sathyanarayana et al., 2024), they recommended 200 observations as the minimum sample size for the implementation of a multivariate analysis in social scientists. This sample size is thought to be adequate to generate reliable parameter estimates, lower sampling error, and improve statistical testing's robustness. Given the number of variables included in this study, a sample size of at least 200 is believed to be sufficient when attempting to conduct statistical testing and test hypotheses.

Sampling Technique

This research uses a non-probability sample, that is purposive sampling. Purposive sampling refers to the selection of respondents in accordance with pre-set criteria related to the subject matter, or the research goals (Herlina et al., 2025). In that case, the respondents here should be Malaysian consumers who have ever used AI chatbots in online customer service. This method is appropriate as it requires participants to have the requisite knowledge or experience to offer informed answers. Purposive sampling is quite commonly adopted in technology acceptance and the online consumer behavior type studies which need special user experience (Herlina et al., 2025).

3.6 Data Collection Method

Data collection is the gathering of systematic information which is necessary for answering a research problem and testing hypotheses (Bhandari, 2020). Secondary data is applied in this research, but also primary data is used to obtain a complete understanding of consumer trust towards AI chatbot in online customer service in Malaysia. The utilization of multiple sources confirms the validity of the construct adding support to the theoretical underpinnings for the study and credibility to its design.

3.6.1 Primary Data

The research instrument of this study is a structured questionnaire that will be distributed by Microsoft Forms. During primary data collecting phase, it provides an opportunity to gather first-hand information that is directly related to the objectives and hypotheses of the study, meaning that the collected data will be relevant and context specific.

The target of this study focuses on students Multimedia University (MMU) mainly from Faculty of Business (FOB), Faculty of Information Science and Technology (FIST), Faculty or Engineering and Technology (FET) and Faculty or Law (FOL). The students have also been regarded as suitable respondents allowing that they are heavy users of digital technologies, online platforms as well as e-commerce services, which means that they may or will have experienced an interaction with AI chatbots for online customer service before. University students are often the respondents of choice for studies involving technology adoption and AI technologies, because they use the digital systems regularly, and therefore can give expert evaluations (Sustaningrum & Haldaka, 2025).

The survey will include Likert scale questions which will measure perceived usefulness, perceived ease of use, perceived interactivity, user trust and anthropomorphism towards chatbots. All items of the measure will be derived from existing and validated scales reported in literature to assure content validity. Responses will be recorded using a Likert scale, commonly used in behavior and technology acceptance research to quantitatively evaluate attitudes and perceptions (Koo & Yang, 2025).

The Microsoft Forms generated questionnaire link can be uploaded online via the MMU student communication system such as outlook, and personal WhatsApp groups. Online survey dissemination is appropriate for this research because it enables easy access to the

respondents in university setting, lowers the cost of collecting data and faster responses (Sustaningrum & Haldaka, 2025). In addition, Microsoft Forms allows for automatic data processing which reduces the occurrence of data entry mistake and helps promote its accuracy.

Description of a pilot study before the full large-scale survey is done, students at MMU can test out the questionnaire on a small group of students to test for how clear, reliable and comprehensible are the questions. Changes suggested will be incorporated. Ethical issues such as voluntary participation, informed consent and confidentiality of respondents' information will be adhered to during the data collection process (Mirza et al., 2023).

3.7 Questionnaire Development

Development of the questionnaire is a systematic procedure to ensure an instrument that adequately measures respondents' perceptions, attitudes and experiences in relation to research variables (Yusoff et al., 2021). In this research, the questionnaire is developed in accordance with the study objectives along with its conceptual framework and hypotheses that have been formulated from the previous chapters. The items of measurement will be developed based on existing and validated scales in previous research on AI chatbots, technology acceptance as well as consumer trust, so that the content validity can be achieved. The questionnaire comprises sections such as respondent profile, behavioural profile, independent variable and dependent variable. All item measures by closed-ended questions to be suitable for statistical analysis.

Demographic Variable

Demographic variables allow researchers to consider respondent characteristics that might affect perceptions and beliefs, providing context for the findings (Yusoff et al., 2021). In this research, within the demographic variables that serve as a description of the respondent and related to characteristics ensuring conditioning factors for application in its context in the research (Yusoff et al., 2021). The choice of demographic variables also reflects the technology adoption and consumer behavior contexts, as well as address trust towards AI chatbots. The studied demographic factors are gender, age, faculty and the level of education. Gender and age are added as they might affect technology use behavior and trust perceptions. Faculty and

study year are added to represent those participants' academic profile and the extent that they had been exposed to tech knowledge.

Table 3. 1: Demographics Variables

No	Proposed Question	Scale	Source
1.	Gender	1- Male 2- Female	(Ding & Najaf, 2024)
2.	Age	1- 18-20 2- 21-23 3- 24-27 4- 28-30	(Ding & Najaf, 2024)
3.	Faculty	1- FOB 2- FIST 3- FET 4- FOL	--
4.	Programme	1- Banking and Finance 2- Human Resource Management 3- International Business 4- Marketing Management 5- Digital Business Management 6- Data Communications and Networking 7- Business Intelligence and Analytics 8- Artificial Intelligence 9- Security Technology 10- Bioinformatics 11- Electronic Engineering 12- Mechanical Engineering 13- Law 14- Other	--
5.	Education Level	1- Foundation 2- Diploma 3- Degree 4- Postgraduate	(Alagarsamy & Mehrolia, 2023)

Behavioral Variable

Behavioral variables reflect the history and pattern of interaction between respondents and AI chatbots (Yusoff et al., 2021). Behavioural factors are critical because when people trust depends on frequency of use, familiarity and context of use (Yusoff et al., 2021). Users with richer AI chatbot experience are more qualified to rate how well chatbots perform and gain trust. Behavior variable are experience, usage frequency and purpose of using chatbot use in internet customer service context were based on this study (Yusoff et al., 2021). These are control variables that can make sure the respondents have relevant exposure with AI chatbots, and researcher may understand trust perception more accurately.

Table 3. 2: Behavioural Variables

No/Code	Proposed Question	Scale	Source
BV1	Have you ever interacted with an AI chatbot for online customer service?	Yes / No	(Prakash et al., 2023)
BV2	How frequently do you use AI chatbots for online customer service?	Never / Less than once a month / 1–3 times a month / 1–2 times a week / Daily	(Ding & Najaf, 2024)
BV3	For what purpose do you usually use AI chatbots?	Product inquiry / Order tracking / Complaints / Payment or refund inquiries / Technical support / Other	(Prakash et al., 2023)
BV4	Which platform do you usually use to interact with AI chatbots for online customer service?	E-commerce websites (e.g., Shopee, Lazada) / Retail or service mobile applications (e.g., banking apps, airline apps) / Social messaging applications (e.g., WhatsApp, Facebook Messenger, Telegram) / Company official websites (live chat/chatbot feature) / Email-based customer service with automated responses	(Adam et al., 2021)

3.7.1 Dependent Variable

Consumer trust in AI chatbots is the extent to which an AI chatbot's ability, dependability, and honesty may be re-lied upon and trusted by users in online service interactions with an average level of trust (Prakash et al., 2023; Rafiq et al., 2022). In this particular research, the DV, consumer trust in AI chatbots, is the primary output construct influenced by perceived usefulness, perceived ease of use, perceived interactivity, perceived risk and anthropomorphism. The consumer trust measurement items are based on previous empirical research into trust in AI and online service systems. All items are rated on a seven-point Likert scale of 1 = Strongly Disagree to 7 = Strongly Agree.

Table 3. 3: Dependent Variable

Code	**Original Question**	**Adapted Question**	**Source**
DV1	I trust that AI will recommend the best services for my needs and demands.	I trust AI chatbots to provide reliable customer service.	(Chi & Hoang Vu, 2023)
DV2	I trust that people are safe when interacting with AI	I feel confident using AI chatbots for online customer service.	(Chi & Hoang Vu, 2023)
DV3	I trust that AI will recommend the best services for my needs and demands	AI chatbots act in my best interest when handling my requests.	(Chi & Hoang Vu, 2023)
DV4	I trust that people are safe when interacting with AI.	AI chatbots provide customer service in a trustworthy manner.	(Chi & Hoang Vu, 2023)

3.7.2 Independent Variable I: Perceived Usefulness

Perceived Usefulness indicates the users' belief that the intelligence-integrated chatbot can deliver accurate information, resolve issues effectively and make their online shopping experience better (Prakash et al., 2023; Rafiq et al., 2022). Perceived usefulness is a major construct of the TAM and has been extensively used in research related to e-commerce, digital services and AI systems (Prakash et al., 2023). In the present study, Perceived Usefulness is considered as an independent variable, since a user will accept AI Chatbots if they perceive it to be of use and adding value in terms of performance (Alagarsamy & Mehrolia, 2023; Rafiq et al., 2022). Measurement Items derived from validated scale in the previous studies that are measured using five- point Likert scales.

Table 3. 4: Independent Variable I: Perceived Usefulness

Code	**Original Question**	**Adapted Question**	**Source**
PU1	Using chatbot improves my performance in seeking customer service support from the provider	AI chatbots help me resolve customer service issues efficiently.	(Prakash et al., 2023)
PU2	The chatbot provides customers with specific, preferred information	AI chatbots provide useful information for my online shopping needs.	(Rafiq et al., 2022)
PU3	Searching with assistance from the chatbot saves me time	AI chatbots save my time when seeking customer service support.	(Rafiq et al., 2022)
PU4	I find the chatbots to be useful for getting customer service queries answered	AI chatbots are useful for online customer service.	(Prakash et al., 2023)

3.7.3 Independent Variable II: Perceived Ease of Use

Perceived ease of use is the extent to which a user feels that using an AI chatbot is effortless (Hoo et al., 2023). When talking about AI chatbot context, this concept relates to how easy to use and intuitive the chatbot system (Alagarsamy & Mehrolia, 2023). The perception of ease of use is being considered as an independent variable since easier systems can reduce the users' cognitive effort, and the level of doubt experienced by them, subsequently influencing trust formation (Hoo et al., 2023). Ease of use is particularly relevant for our student population who are regular users of digital platforms, and it serves to influence their initial perceptions of trust. The items of measurement are from the previous studies based on TAM.

Table 3. 5: Independent Variable II: Perceived Ease of Use

Code	**Original Question**	**Adapted Question**	**Source**
PEU1	My interactions with the chatbot were clear and understandable	It is easy for me to interact with AI chatbots for customer service.	(Prakash et al., 2023)
PEU2	The chatbot is easy to use	I find AI chatbots easy to understand and use.	(Prakash et al., 2023)
PEU3	Learning to navigate through e-commerce websites is simple with assistance from the chatbot	Learning to use AI chatbots is easy for me.	(Rafiq et al., 2022)
PEU4	Using chatbots for tourism is enjoyable	AI chatbots are user-friendly.	(Rafiq et al., 2022)

3.7.4 Independent Variable III: Perceived Interactivity

Perceived interactivity is described as how users perceive the responsiveness, interaction, and two-way communication of an AI chatbot (Ding & Najaf, 2024; Rafiq et al., 2022). In online customer service, perceived interactivity represents the degree to which the chatbot does a good job of responding to user inquiries, keeping the conversation moving on pace, as well as serving users' unique needs (Prakash et al., 2023). This predictor is based on the expectation that more interactional chatbots can generate more enjoyable and meaningful interactions that influence a user's perceived control and trust (Alagarsamy & Mehrolia, 2023; Fu et al., 2024). In AI-driven interfaces, interactivity is a bridge between technical affordances and social existence (Ding & Najaf, 2024; Rafiq et al., 2022). Measurement items are derived from existing research on HCI and AI chatbot.

Table 3. 6: Independent Variable III: Perceived Interactivity

Code	**Original Question**	**Adapted Question**	**Source**
PI1	AI answers my requests quickly	AI chatbots respond promptly to my questions.	(Chi & Hoang Vu, 2023)
PI2	The Chatbot enabled two-way communication	AI chatbots allow two-way communication during service interactions.	(Prakash et al., 2023)
PI3	I felt that the Chatbot carefully registered my responses and gave feedback based on the information I entered	AI chatbots provide relevant responses during interaction.	(Prakash et al., 2023)
PI4	Chatbot had the ability to respond to my specific needs quickly	I feel involved when interacting with AI chatbots.	(Rafiq et al., 2022)
PI5	My interaction with the Chatbot felt like a continuous thread	AI chatbots maintain a smooth and continuous conversation.	(Prakash et al., 2023)

3.7.5 Independent Variable IV: Perceived Risk

Perceived risk relates to the same pattern of thoughts of how much users believe in the vulnerability they face with AI chatbots leading toward undesirable consequences, such as loss of privacy, data misappropriation or wrong information (Bouhia et al., 2022; Li et al., 2023). Perceived risk is one of the major considerations in e-commerce especially so when personal and transaction related information is being exchanged (Gupta et al., 2023). The greater the perceived risk, lower trust on digital and AI based services might result. By explaining how students perceive risk, the pattern of concerns for trust in using a chatbot can be clarified (Bouhia et al., 2022; Li et al., 2023). Measurement items are based on the previous studies of online risk and AI trust.

Table 3. 7: Independent Variable IV: Perceived Risk

Code	**Original Question**	**Adapted Question**	**Source**
PR1	I am worried to use chatbot because other people or organizations may be able to access my account	I am concerned about my personal data security when using AI chatbots.	(Prakash et al., 2023)
PR2	My personal information could be misused	Using AI chatbots involves privacy risks.	(Bouhia et al., 2022)
PR3	There will be much potential loss associated with disclosing personal information to the chatbot	I fear potential harm from my data being shared with third parties through AI chatbots.	(Prakash et al., 2023)
PR4	I did not know exactly what to expect	I feel uncertain about the outcomes of using AI chatbots.	(Bouhia et al., 2022)
PR5	I am worried to use chatbot because other people or organizations may be able to access my account	I worry about negative consequences when using AI chatbots.	(Prakash et al., 2023)

3.7.6 Independent Variable V: Anthropomorphism

Anthropomorphism is the degree to which users believe that an AI chatbot includes human-like attributes like friendliness, empathy, and conversational ability (Li et al., 2023; Rafiq et al., 2022). Anthropomorphic appearances can decrease the psychological distance between human users and AI systems, generating an appearance of greater naturalness and social presence in interactions (Chi & Hoang Vu, 2023). This variable is added as the human-like characteristics of chatbots may help to increase emotional connection, social presence and trust especially in customer service settings (Fu et al., 2024; Prakash et al., 2023). Although the level of anthropomorphism must be tempered to not encourage unrealistic expectations (Shahbandi, 2025). The measurement items are modified from previous research on the anthropomorphism of chatbots.

Table 3. 8: Independent Variable V: Anthropomorphism

Code	**Original Question**	**Adapted Question**	**Source**
ANT1	Conversations with chatbots should be natural	AI chatbots communicate in a human-like manner.	(Rafiq et al., 2022)
ANT2	AI application has friendly action	AI chatbots appear friendly during interaction.	(Chi & Hoang Vu, 2023)
ANT3	Chatbots should seem as if they understand the person with whom they are interacting	AI chatbots show understanding of my needs.	(Rafiq et al., 2022)
ANT4	It is important that the conversation with a chatbot resembles one with a human being	Interacting with AI chatbots feels similar to talking to a human.	(Rafiq et al., 2022)

3.8 Data Analysis Method

Data analysis is the process of inspecting, cleaning, transforming, and modelling data with the goal of discovering useful information for decision-making purposes (Alem, 2020). The purpose of data analysis for this study is to describe the characteristics of respondents and explore the relationships among variables as well as differences between groups, thus investigating possible hypotheses based on conceptual framework (Alem, 2020).

For this study, two types of data analysis will be performed: descriptive and inferential (hypothesis testing). Descriptive and inferential analyses are performed to summarize the characteristics of respondents; to describe key variables; as well as to analyse the relationships between independent and dependent variables. All analyses will use SPSS version 30.0; a software that is commonly used by academic researchers in fields of social science and business research because of its reliability and simple analysis process.

3.8.1 Descriptive Analysis

Descriptive analysis is used to summarize and describe basic features of the data critically evaluate the empirical evidence on what other factors influenced whether a model was used (Alem, 2020). Descriptive analysis is performed to describe a summary of respondent profiles, behavioural characteristics and the general response patterns for each variable in this study (Alem, 2020). Such analysis is useful to illustrate the distribution, averages and range of instrument data prior to inferential testing.

Descriptive statistics will be used in this study including frequency, percentage, mean and standard deviation. Frequency and percentage will be used for demographic variables (gender, age group, faculties, levels of study) and behavioural variables (usage experience with AI chatbot(s), frequency of usage, reasons for its/their use) in terms of describing the profile of respondents. Responses to the independent variables (perceived usefulness, perceived ease of use, perceived interactivity, perceived risk and anthropomorphism) as well as for the dependent variable (consumer trust) will be summarized with means and standard deviations. These analytic methods can give perspective to the general attitudes of respondents and the variability in responses among constructs.

3.8.2 Hypothesis Testing

Hypotheses testing is an inferential statistical technique that attempts to determine whether there is enough evidence in a sample of data to support a belief or claim about the value of hypothesized relationships (Alem, 2020). The purpose of hypothesis testing in this study is to assess the existence and strength of connections among independent variables and dependent variable and thus verify or confirm research framework and theoretical assumptions behind the study.

Pearson correlation analysis and multiple regression analysis will be used in empirically testing the hypotheses. Pearson's correlation analysis will be employed to assess the relationship between and each of independent variable (perceived usefulness, perceived ease of use, perceived interactivity, perceived risk and anthropomorphism) and dependent variable (consumer trust). This technique applies to how covariation among continuous variables, which are both measured by Likert scales, is assessed.

The combined and relative effects of the impact dimensions on consumer trust will then be analysed by regression analysis. Such analysis enables the researcher to know which independent variables make a significant contribution to predicting out consumer trust, after accounting for effect of other variables (Kwak, 2023). A $p < 0.05$ level of significance is availed to determine whether the null hypotheses will be accepted or rejected (Kwak, 2023). The findings from these analyses will empirically test the hypotheses and contribute to a broader understanding of what determines consumer trust in AI chatbots for online customer service in Malaysia.

3.9 Summary

This chapter has discussed the research methods applied in examining factors influencing consumer trust in AI chatbots for online customer service among Malaysian e-commerce. A structured questionnaire was used to provide for systematic data collection and objective measurement of the study variables with a quantitative research design. The chapter described the research design, development of hypothesis, sample design instrument and data collection procedure, instrument pre-testing and final administration and techniques of analysing the data collected for study.

The extraction technique introduced in this chapter is a strong basis for analysing the data (Chapter 4). The application of validated measures and a Likert-scale questionnaire allows for valid descriptive and inferential statistical assessment. Moreover, the sampling approach and data collection method used in this study guarantee data validity and representativeness of target samples for meaningful findings interpretation.

The procedures, such as descriptive statistics, correlation analysis and multiple regression model established, through which this study can systematically test the research hypotheses and analyse associations of the independent variables and consumer trust. As a result, the research method helps to interpret data reliably and empirically validate the research model which will be introduced and discussed in chapter 4.

CHAPTER 4

DATA ANALYSIS

4.1 Overview

This chapter shows the analysis and result of data collected from this study regarding factors that influence consumer to trust on AI Chatbot online customer service in Malaysia. The questionnaire surveys data were analysed with Statistical Package for the Social Sciences (SPSS) Version 30.0. This chapter aims to analyse the connections between independent variables (perceived usefulness, perceived ease of use, perceived interactivity, perceived risk attentiveness and anthropomorphism) and the dependent variable, consumer trust in AI chatbots.

This chapter commences with the demographic and behavioural profiles of the respondents, and that precedes a descriptive analysis to provide a summary of the generic features that characterize what has been collected in the data. Then, a reliability analysis is performed to evaluate the internal consistency of measurement items. Followed by that, inferential statistical analyses such as multiple regression analysis is developed to examine the proposed hypotheses to check how significant relationships tend to define between variables. The results in this chapter also provide empirical evidence for the research objectives and research questions of the study.

4.2 Normality Test

All quantitative studies must complete a normality test before conducting any follow-up statistical analysis, to verify that their data meets the normal distribution assumption; this study uses skewness and kurtosis to assess the normality of all measurement items. Skewness measures the symmetry of a distribution, while kurtosis measures the peakedness or flatness of a distribution (Kim, 2013).

In academia, the universally accepted normality qualification thresholds for social science research are a skewness range of -2 to +2 and a kurtosis range of -7 to +7 (Wah, 2023). The skewness values measured in this study range from -1.515 to -1.016, and the kurtosis values range from -0.420 to 1.475; all data fully comply with the normality requirements.

For all measured variables in this study, their negative skewness values show that respondents' feedback overall tends toward higher levels of agreement. No abnormal extremes of excessive peakedness or flatness appeared in the kurtosis values. The data approximately follows a normal distribution, making it suitable for carrying out correlation analysis and multiple regression analysis. All skewness and kurtosis values are presented in Table 4.1.

Table 4. 1: Skewness and Kurtosis

	Skewness	**Kurtosis**
CT1	-1.425	1.047
CT2	-1.306	0.583
CT3	-1.405	1.089
CT4	-1.499	1.338
PU1	-1.273	0.719
PU2	-1.223	0.632
PU3	-1.288	0.987
PU4	-1.415	1.156
PEU1	-1.164	0.212
PEU2	-1.355	0.987
PEU3	-1.053	0.141
PEU4	-1.161	0.086
PI1	-1.213	0.609
PI2	-1.163	0.302
PI3	-1.016	-0.006
PI4	-1.171	0.275
PI5	-1.030	-0.420
PR1	-1.225	0.600
PR2	-1.456	1.469
PR3	-1.319	0.991
PR4	-1.305	0.849
PR5	-1.515	1.475
ANT1	-1.200	0.488
ANT2	-1.326	0.927
ANT3	-1.060	0.226
ANT4	-1.210	0.382

4.3 Reliability Test

This study conducted reliability analysis to test the internal consistency of the measurement items used in the research. Cronbach's α coefficient was adopted to assess the reliability level of each research construct. Following widely accepted academic norms, an α value ≥0.70 was deemed qualified, and an α value >0.80 was deemed good (Ahmad et al., 2024).

To ensure the validity and consistency of measurement items in measuring their corresponding constructs, this study only included the items retained from the final factor analysis in the reliability analysis. The reliability test covers for the dependent variable, Consumer Trust in AI Chatbots (DV), and the five independent variables, which is Perceived Usefulness (IV1), Perceived Ease of Use (IV2), Perceived Interactivity (IV3), Perceived Risk (IV4), and Anthropomorphism (IV5).

In this study, the Cronbach's α coefficients of all measured constructs were higher than 0.8, with values ranging from 0.843 to 0.922. Their internal consistency reached a level from good to excellent. Specifically, Anthropomorphism (IV5) had the greatest dependability rating of 0.922, followed by Consumer Trust (DV) with 0.898 and Perceived Risk (IV4) with 0.889. Perceived Usefulness (IV1) and Perceived Interactivity (IV3) both obtained reliability values of 0.873, while Perceived Ease of Use (IV2) recorded the lowest value of 0.843, which still met the requirement for good reliability.

All constructs included in this study passed the reliability and internal consistency tests, so the measurement items used are suitable for the subsequent correlation and regression analyses.

Table 4. 2: Summary of Cronbach's Alpha values and number of items

	Cronbach Alpha	**No. of items**
Consumer Trust	0.898	4
Perceived Usefulness	0.873	4
Perceived Ease of Use	0.843	4
Perceived Interactivity	0.873	5
Perceived Risk	0.889	5
Anthropomorphism	0.922	4

4.4 Demographic Profiling

In this study, we developed a demographic profile to describe the background characteristics of respondents. We adopted descriptive analysis centered on frequency and percentage to process the demographic information collected via questionnaires, included five core variables such as gender, age, faculty, programme, and education level and presented the results in tables and figures to enhance the clarity of the profile.

Table 4. 3: Demographic Profiling

Demographic Variable	**Frequency**	**Percent (%)**
Gender		
Male	109	45.4
Female	131	54.6
Total	240	100.0
Age		
18-20	53	22.1
21-23	149	62.1
24-27	23	9.6
28-30	15	6.3
Total	240	100.0
Faculty		
FOB	112	46.7
FIST	80	33.3
FET	30	12.5
FOL	18	7.5
Total	240	100.0
Programme		
Banking and Finance	14	5.8
Human Resource Management	10	4.2
International Business	18	7.5
Marketing Management	25	10.4
Digital Business Management	28	11.7
Data Communications and Networking	10	4.2
Business Intellignece and Analytics	11	4.6

Artificial Intelligence	27	11.3
Security Technology	13	5.4
Bioinformatics	8	3.3
Electronics Engineering	8	3.3
Mechanical Engineering	21	8.8
Law	18	7.5
Other	29	12.1
Total	240	100.0
Education Level		
Foundation	23	9.6
Diploma	32	13.3
Degree	174	72.5
Postgraduate	11	4.6
Total	240	100.0

4.4.1 Gender

This paper draws on the survey data from Table 4.3. Among the 240 respondents of this research, 131 are female, accounting for 54.6% of the total sample, and 109 are male, accounting for 45.4%. Although female respondents form the majority of the sample, the gap between the proportion of male and female respondents is small, and both groups are fully represented in the study.

4.4.2 Age

The age distribution of the samples in this study is detailed in Table 4.3, sorted in descending order of group proportion: the 21–23 year-old group has 149 participants, accounting for 62.1%; the 18–20 year-old group has 53 participants, accounting for 22.1%; the 24–27 year-old group has 23 participants, accounting for 9.6%; and the 28–30 year-old group has 15 participants, accounting for 6.3%.

Most respondents in this study are young adults, which matches the pre-set target group of college students who actively use digital technologies, e-commerce platforms, and AI chatbots. Therefore, the sample is appropriate for this study's assessment of consumer trust in online customer service AI.

4.4.3 Faculty

This study relies on Table 4.3 to present the sample distribution of respondents by their affiliated college, sorted in descending order of proportion: FOB (Faculty of Business), 112 respondents accounting for 46.7%; FIST (Faculty of Information Science & Technology), 80 respondents accounting for 33.3%; FET (Faculty of Engineering), 30 respondents accounting for 12.5%; FOL (Faculty of Law), 18 respondents accounting for 7.5%.

This paper presents an empirical study focused on consumers' trust in AI chatbots in the online customer service context. The study's sample consists mainly of respondents from business and technology academic departments, a group that regularly interacts with relevant technologies and is able to provide the valid insights required for this research.

4.4.4 Programme

The data on the distribution of academic programs among respondents of this survey is sourced from Table 4.3. Respondents come from a wide range of academic programs, according to Table 4.6. "Other" had the most respondents (29, or 12.1%), followed by "Digital Business Management" (28, or 11.7%) and "Artificial Intelligence" (27, or 11.3%). There were 25 responders (10.4%) in marketing management and 21 respondents (8.8%) in mechanical engineering.

Furthermore, "International Business" and "Law" each had 18 participants, accounting for 18 or 7.5% of the total sample, followed by 14 respondents (5.8%) from "Banking and Finance" and 13 respondents (5.4%) from "Security Technology". "Human Resource Management" and "Data Communications and Networking" both had 10 respondents (4.2%). "Bioinformatics" and "Electronic Engineering" each had 8 participants, accounting for 3.3%, making them the two majors with the smallest sample sizes across all disciplines.

The respondents in this study come from diverse academic backgrounds including business, technology, engineering, and law. The diversity of the study sample enhances the research's representativeness and broadens perspectives for studies on consumer trust in AI chatbots for online customer service in Malaysia.

4.4.5 Education Level

The educational background structure of the valid survey samples collected for this study is detailed in Table 4.3. Of all samples, 174 are degree students, accounting for 72.5%; 32 are diploma students, accounting for 13.3%; 23 are foundation students, accounting for 9.6%; and only 11 are postgraduate students, accounting for 4.6%.

This study uses Malaysia's e-commerce sector as its research context, evaluating the public's perceptions of and trust in AI chatbots for online customer service. The study's research sample has a high level of exposure to digital services, which meets the requirements for eligible respondents, and can rule out doubts regarding the sample's applicability for this research.

4.5 Behavioral Analysis

This study conducts behavioral analysis of respondents who use online customer service AI chatbots. It applies descriptive statistical methods using frequency and percentage and incorporates four categories of core behavioral variables. The conclusions drawn from this study can supplement relevant industry insights into respondents' familiarity with and exposure to AI technologies.

Table 4. 4: Behavioral Analysis

Behavioral Variable	Frequency	Percent (%)
BV1: Prior Use		
Yes	240	100.0
No	0	0
Total	240	100.0
BV2: Usage Frequency		
Never	2	0.8
Less than once a month	21	8.8
1-3 times a month	26	10.8
1-2 times a week	82	34.2
Daily	109	45.4
Total	240	100.0
BV3: Usage Purpose		
Product inquiry	61	25.4
Order tracking	60	25.0
Complaints	32	13.3
Payment or refund inquiries	37	15.4
Technical support	33	13.8
Other	17	7.1
Total	240	100.0
BV4: Preferred Platform		
E-commerce websites (e.g., Shopee, Lazada)	62	25.8
Retail or service mobile applications (e.g., banking apps, airline apps)	21	8.8

Social messaging applications (e.g., WhatsApp, Facebook Messenger, Telegram)	80	33.3
Company official websites (live chat/chatbot feature)	44	18.3
Email-based customer service with automated responses	24	10.0
Other	9	3.8
Total	240	100.0

4.5.1 BV1: Experience Interacting with AI Chatbots for Online Customer Service

Data from Table 4.4 of this study shows that all 240 respondents have experience using AI customer service chatbots, and all are qualified samples suitable for this study. Currently, AI chatbots have become widely popular among consumers, and are primarily used for the online customer service scenarios of e-commerce platforms and digital applications (Gao et al., 2025); all Malaysian respondents in this study have prior usage experience and are able to provide reliable evaluations of their level of trust in the relevant services.

4.5.2 BV2: Frequency of Using AI Chatbots for Online Customer Service

This study uses the respondent sample data from the designated survey form Table 4.4 to compile statistics on the user usage frequency of AI online customer service chatbots for e-commerce and digital platform scenarios in Malaysia. The frequencies are sorted from highest to lowest as follows: 109 respondents use the service daily, accounting for 45.4%; 82 respondents use it 1-2 times per week, making up 34.2%; 26 respondents use it 1-3 times per month, accounting for 10.8%; 21 use it less than once per month, making up 8.8%; and 2 respondents have never used it, accounting for 0.8%.

Most respondents engage in high-frequency interactions and have a high level of familiarity with AI customer service. This proportion of usage also reflects that the reliance of local online services on AI chatbots continues to rise.

4.5.3 BV3: Purpose of Using AI Chatbots

Based on the data in Table 4.4 collected for this paper's research, respondents' uses of AI chatbots are arranged in descending order of their proportion. The use with the highest share is product inquiry, which involves 61 respondents, accounting for 25.4% of the total, followed by order tracking coming in second with 60 respondents (25.0%). 37 respondents (15.4%) had questions about payments or refunds, while 33 respondents (13.8%) had complaints and 32 respondents (13.3%) inquire for technical support. Furthermore, 17 respondents (7.1%) indicated "other" purpose for using AI chatbots.

Findings from this study's survey show that the participating consumers primarily use AI chatbots in e-commerce scenarios to obtain product information and track orders. Both functions are universal platform-wide customer service features that can broadly provide rapid assistance for routine customer inquiries.

4.5.4 BV4: Platform Used to Interact with AI Chatbots for Online Customer Service

This study relies on the survey samples presented in Table 4.4. The proportion of respondents accessing AI chatbots through different channels, ranked from highest to lowest, are social and communication apps (33.3%), e-commerce websites (25.8%), and corporate official websites equipped with chatbot functions (18.3%). Statistics collected in this survey show that 24 respondents, accounting for 10% of all participants, have used auto-reply email customer service; 21 respondents, or 8.8%, have used embedded AI chatbots from airlines and banks; and only 9 respondents, making up 3.8%, have used such services on other platforms.

This study finds that Malaysian consumers most frequently interact with AI chatbots via social messaging apps and e-commerce websites. This trend not only reflects users' preference for convenient online customer service channels but also demonstrates that the level of AI integration across these two types of local platforms is continuously rising.

4.6 Descriptive Analysis

This study will conduct descriptive statistical analysis to examine the mean and standard deviation of all included variables. Descriptive analysis is a category of statistical methods that can systematically summarize and describe the core characteristics of a dataset, enabling us to sort out the overall patterns in respondents' answers (Siregar et al., n.d.). In this study, the mean represents the average level of respondents' agreement with each measurement item, while the standard deviation represents the degree of dispersion of the responses. This study only includes items retained after reliability analysis, and the results will be presented sequentially for each variable within the research framework.

4.6.1 Consumer Trust in AI Chatbots

This study relies on the survey data from Table 4.5. The overall mean trust score of respondents for online customer service AI chatbots is 5.73, indicating an overall high level of trust. Among the individual survey items, DV4 has the highest mean value of 5.81 (SD=1.661), while DV2 has the lowest mean value of 5.62 (SD=1.798), and the latter still falls within the range of positive agreement.

This study observed that respondents' average perception of AI chatbots is relatively high, which may stem from their frequent exposure to similar tools on e-commerce and digital platforms, the fact that most respondents are active users of online customer service, and their general belief that such tools are reliable and practical.

Table 4. 5: Descriptive Statistics for Consumer Trust in AI Chatbots.

Item	**Mean**	**Std. Deviation**
DV1	5.72	1.715
DV2	5.62	1.798
DV3	5.76	1.623
DV4	5.81	1.661
Total	5.73	1.699

4.6.2 Perceived Usefulness

This study draws on the survey statistical table Table 4.6 to report data related to the perceived usefulness (PU) dimension of AI chatbots in the online customer service scenario: the overall mean of PU is 4.12; within this dimension, PU4 records the highest mean (M=4.17, SD=1.138), while PU1 has the lowest mean (M=4.06, SD=1.174). Respondents generally recognized the practicality of the chatbots, with stable, consistent cognitive evaluations. Their core values fall into three categories: improving efficiency, enabling quick access to information, and resolving customer problems.

Table 4. 6: Descriptive Statistics for Perceived Usefulness

Item	**Mean**	**Std. Deviation**
PU1	4.06	1.174
PU2	4.12	1.107
PU3	4.12	1.084
PU4	4.17	1.138
Total	4.12	1.126

4.6.3 Perceived Ease of Use

This study analyzes the core variable perceived ease of use (PEU) based on Table 4.7. This variable has an overall mean of 4.05, showing that respondents generally thought AI chatbots were simple to use and engage with and most respondents recognize their ease of use. PEU2 has the highest mean score, while PEU4 has the lowest mean score and a larger standard deviation, which signals that a small number of respondents hold divergent perceptions of this dimension.

Table 4. 7: Descriptive Statistics for Perceived Ease of Use

Item	**Mean**	**Std. Deviation**
PEU1	4.06	1.212
PEU2	4.11	1.161
PEU3	4.02	1.160
PEU4	3.99	1.318
Total	4.05	1.213

4.6.4 Perceived Interactivity

Based on the survey data from Table 4.8, Perceived Interactivity (PI) has an overall mean value of 3.99. Respondents generally recognize the interactivity of AI chatbots; among all items in the measurement scale, PI1 scores the highest (M=4.09, SD=1.131), while PI5 scores the lowest (M=3.90, SD=1.428).

The respondents of this study generally believe that this AI customer service chatbot meets the required standards for its response capability and can maintain normal online interactions; however, the higher standard deviation of the PI5 indicator shows that respondents hold markedly divergent evaluations of its human-like communication ability.

Table 4. 8: Descriptive Statistics for Perceived Interactivity

Item	**Mean**	**Std. Deviation**
PI1	4.09	1.131
PI2	4.07	1.171
PI3	3.91	1.239
PI4	4.00	1.259
PI5	3.90	1.428
Total	3.99	1.246

4.6.5 Perceived Risk

According to statistics from Table 4.9, the overall mean of perceived risk in this study is 4.14. The mean and standard deviation of PR2 is 4.21 and 1.059 respectively which scores the highest mean, while those of PR3 are 4.08 and 1.145 respectively which scores the lowest mean.

This study's survey finds that all respondents are aware that AI chatbots pose risks related to privacy, misinformation, and data security, yet they still maintain high levels of trust in these tools and use them at high frequencies, as their perceived benefits outweigh these associated risks.

Table 4. 9: Descriptive Statistics for Perceived Risk

Item	**Mean**	**Std. Deviation**
PR1	4.11	1.126
PR2	4.21	1.059
PR3	4.08	1.145
PR4	4.14	1.125
PR5	4.15	1.169
Total	4.14	1.125

4.6.6 Anthropomorphism

Data from Table 4.10 of this study indicates that the overall mean of the anthropomorphism dimension is 4.01. Among the individual survey items, ANT2 scores the highest (M=4.06, SD=1.191) while ANT4 scores the lowest (M=3.95, SD=1.280). The vast majority of respondents perceive that AI chatbots possess human-like characteristics.

This study's survey found that respondents moderately agreed that AI chatbots possess three categories of anthropomorphic communication traits. However, the average agreement score for this view was lower than the mean values of all other variables measured in the study, and respondents generally still believed that AI chatbots cannot fully replicate human communication or achieve sufficient emotional understanding.

Table 4. 10: Descriptive Statistics for Anthropomorphism

Item	**Mean**	**Std. Deviation**
ANT1	4.05	1.179
ANT2	4.06	1.191
ANT3	3.96	1.174
ANT4	3.95	1.280
Total	4.01	1.206

4.7 Multiple Linear Regression

Multiple linear regression is a class of statistical methods that can predict the value of a single dependent variable based on the values of multiple independent variables. The core premise for this method to be valid is that a linear correlation exists between the variables (Uyanık & Güler, 2013). This study focuses on the online customer service scenario and adopts multiple linear regression analysis. It sets perceived usefulness, perceived ease of use, perceived interactivity, perceived risk, and anthropomorphism as independent variables, and consumers' trust in AI chatbots as the dependent variable. This study aims to verify the explanatory power of the research model and evaluate the predictive effect of the independent variables.

4.7.1 Overall Model summary

Table 4. 11: Model Summary

Model	**Values**
R	0.898
R Square	0.806
Adjusted R Square	0.802
Std. Error of the Estimate	0.66256
Durbin-Watson	1.921

Based on Table 4.11, the R value is 0.898, showing that the data has a very significant positive link between the dependent variable (consumer trust in AI chatbots) and independent variables. The model shows that the R Square value is 0.806, which accounting for 80.6% of the variation in consumer trust can be explained by the five IVs.

The Adjusted R Square value output by the regression model in this study is 0.802. This statistical measure adjusts for interference caused by the number of predictor variables, the model remains robust, and the selected independent variables can strongly explain consumer trust towards AI chatbots for online customer service.

The regression model in this study passed core reliability diagnostics (Arel-Bundock, 2022). Its Durbin-Watson statistic is 1.921, with no severe autocorrelation detected. Furthermore, its standard error of the estimate is 0.66256, confirming the model has good accuracy in estimating consumer trust.

The regression model used in this study meets the required standard for statistical robustness and can support the follow-up analysis of individual effects regarding consumer trust in AI chatbots.

4.7.2 ANOVA Table

Table 4. 12: ANOVA Table

Model	Sum of Squares	df	Mean Square	F	Sig.
Regression	426.952	5	85.390	194.516	<.001[b]
Residual	102.723	234	.439	-	-
Total	529.675	239	-	-	-

This study uses ANOVA analysis. Its core objective is to verify the statistical significance of the regression model that explains the association between independent variables and consumers' trust in AI chatbots, and to test whether this model's predictive performance outperforms the baseline model that includes no predictor variables (Hassan et al., 2024).

This study draws on the test data in Table 4.12, adopts the commonly used statistical threshold of 0.05 in the social sciences, and combined with the results of an F-value of 194.516 and $p<0.001$, determines that the regression model is statistically significant.

The core regression results of this study show that the five selected independent variables jointly produce a significant impact on the dependent variable, namely consumers' trust in online customer service AI chatbots. Existing empirical environmental studies focused on Lake Burullus (Hassan et al., 2024) from the academic community show that a high ratio of the regression sum of squares to the residual sum of squares can effectively prove that the regression model explains most of the variation in the dependent variable.

Furthermore, the regression sum of squares of this study is 426.952, far exceeding the residual sum of squares of 102.723. The model can explain most of the variance in consumer trust, so the model is suitable and can support the subsequent interpretation of individual coefficients and hypothesis testing.

4.7.3 Coefficient Table

Table 4. 13: Coefficient Table

Model	Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Collinearity Statistics	
	B	Std. Error	Beta			Tolerance	VIF
(Constant)	-.062	.200		-.309	.757		
PerceivedUsefulness	.536	.100	.345	5.351	<.001	.200	5.007
PerceivedEaseofUse	-.015	.133	-.010	-.113	.910	.103	9.695
PerceivedInteractivity	.284	.121	.194	2.343	.020	.121	8.273
PerceivedRisk	.165	.105	.104	1.564	.119	.189	5.295
Anthropomorphism	.457	.071	.334	6.434	<.001	.308	3.243

This study conducts statistical analysis centered on the core dependent variable of consumer trust in AI chatbots. It interprets the regression coefficient table, tests the independent effect of each independent variable, as well as the significance level, standardized β coefficients, and multicollinearity statistics of each predictor variable.

Based on the regression results in Table 4.13, this study shows that perceived usefulness has a significant positive correlation with consumer trust ($\beta=0.345$, $p<0.001$). Respondents who view AI customer service as more practical are more likely to trust this service, and usefulness is one of the key factors that influence consumer trust.

The regression results of this study show that the regression coefficient linking Perceived Ease of Use to consumer trust is $\beta = -0.010$, $p = 0.910$, representing a non-significant negative effect. The influence of this variable on trust has been weakened, as the survey respondents were generally familiar with digital technology and AI chatbots.

The regression results of this study show that perceived interactivity is significantly positively correlated with consumer trust ($\beta=0.194$, $p=0.020$). AI chatbots with stronger interactivity are more likely to gain users' trust.

In this study, set in the scenario of AI chatbots, the regression values of perceived risk and consumer trust were measured as $\beta = 0.104$, $p = 0.119$. This result did not reach statistical

significance when applying the 0.05 threshold. It is inferred that the study's respondents paid relatively little attention to relevant risks, due to their high-frequency use of this type of tool.

This study's empirical findings indicate that the anthropomorphism of AI chatbots has a significant positive correlation with consumer trust ($\beta=0.334$, $p<0.001$), confirming that human-like attributes can enhance consumer trust.

According to the data of this study show that perceived usefulness ($\beta=0.345$) recorded the highest coefficient, followed closely by anthropomorphism ($\beta=0.334$); these two factors are the core predictors of consumer trust in AI chatbots.

In this study, the Variance Inflation Factor (VIF) values ranged from 3.243 to 9.695, while the tolerance values ranged from 0.103 to 0.308. Diagnostic tests of the regression model confirmed that all tolerance values were above 0.10 and all variance inflation factor (VIF) values were below 10. No severe multicollinearity was detected, so the regression results are reliable and can be used for subsequent hypothesis testing. The diagnostic values of the present model are within the acceptable range as VIF values are not greater than 10 and tolerance values are not lesser than 0.1 (Senaviratna and Cooray, 2019). Although (Schober & Schwarte, 2018) focused their analysis on correlations, their argument for testing statistical assumptions before interpreting coefficients supports the validity of the diagnostic work applied in this study.

4.8 Hypothesis Testing

This study conducted hypothesis testing to verify the association between the independent variable and consumers' trust in online customer service AI chatbots. Hypothesis testing provides a systematic framework for determining whether an observed effect is real or due to random chance (Su, 2025). We used the p-values output by multiple linear regression to make statistical decisions and set $p < 0.05$ as the judgment threshold for a statistically significant hypothesis.

Table 4. 14: Standardized Regression Coefficients (β) and Significance Testing for Consumer Trust

	Coefficient	**t-value**	**Sig.**	**Result**
PU→ DV	0.345	5.351	<0.001	Significant
PEU→ DV	-0.010	-0.113	0.910	Not Significant
PI→ DV	0.194	2.343	0.020	Significant
PR→ DV	0.104	1.564	0.119	Not Significant
ANT→ DV	0.334	6.434	<0.001	Significant

4.8.1 H1: Perceived Usefulness and Consumer Trust in AI Chatbots

In this study, we propose research hypothesis H1: Perceived usefulness is significantly correlated with consumers' trust in AI chatbots for online customer service. If consumers perceive that the chatbot is useful, efficient, and capable of improving the customer service experience, they are more likely to develop trust in it.

Based on the data from Table 4.14 of this study, perceived usefulness has a standardized coefficient of 0.345, a t-value of 5.351, and $p < 0.001$. With $p < 0.05$ set as the significance threshold, its association with consumer trust is statistically significant.

The primary hypothesis H1 proposed in this study holds true. Respondents' perceived usefulness of AI chatbots can positively promote the establishment of their trust in AI-powered online customer service.

4.8.2 H2: Perceived Ease of Use and Consumer Trust in AI Chatbots

This study proposes Hypothesis H2: In the online customer service scenario, the empathy of AI chatbots will positively enhance consumers' trust in AI chatbots, and the derivation of this hypothesis aligns with the classic logic of the field.

As shown by the statistical data listed in Table 4.14, the standardized coefficient of perceived ease of use is -0.010, its significance value of 0.910 is greater than the 0.05 threshold, and its correlation with consumer trust has no statistical significance.

Empirical tests of this study show that Hypothesis H2 is not supported. In the online customer service context, perceived ease of use has no significant impact on consumers' trust in AI chatbots, and the survey respondents are generally familiar with digital technologies and AI systems.

4.8.3 H3: Perceived Interactivity and Consumer Trust in AI Chatbots

This study proposes Hypothesis H3: There is a significant association between perceived interactivity and consumers' trust in AI chatbots deployed in online customer service scenarios. When a system has responsiveness, interactivity, and the ability to sustain effective communication, consumers are more willing to trust it.

This study draws on data from Table 4.14. Perceived interactivity has a standardized coefficient of 0.194, a t-value of 2.343, and a p-value of $0.020 < 0.05$, and its association with consumer trust is statistically significant.

Hypothesis H3 of this study is supported. Respondents who perceive that the AI chatbots used in online customer service scenarios have interactivity and responsiveness are more likely to develop trust in their services, which highlights the important role of effective interactive communication in enhancing consumer trust.

4.8.4 H4: Perceived Risk and Consumer Trust in AI Chatbots

This paper proposes research hypothesis H4: Consumers' perceived risk is significantly correlated with their trust in online customer service AI chatbots. Perceived risk comprises three categories, namely privacy concerns, security issues, and inaccurate information, and exerts an impact on consumers' trust levels.

Statistical data from Table 4.14 of this study shows that perceived risk has a standardized coefficient of 0.104, a t-value of 1.564, and a significance value of 0.119. Using the 0.05 threshold that is universally adopted in academia, its association with consumer trust is not statistically significant.

Hypothesis H4 of this study was not supported. In the online customer service context, consumers' perceived risk did not significantly affect their trust in AI chatbots. The study's respondents were familiar with the relevant technologies and believed that the risks of using the tool were controllable.

4.8.5 H5: Anthropomorphism and Consumer Trust in AI Chatbots

This study proposes Hypothesis H5: In the online customer service scenario, the anthropomorphism of AI chatbots has a significant correlation with consumer trust; when these chatbots exhibit human-like traits, communication styles and interactive behaviors, they are more likely to win consumers' trust.

The data from Table 4.14 of this study show that the anthropomorphism variable has a standardized coefficient of 0.334, a t-value of 6.434, and $p<0.001$. With $p<0.05$ set as the threshold, its association with consumer trust is statistically significant.

This study verifies that Hypothesis H5 is valid. AI chatbots with anthropomorphic features and an interactive communication style are more likely to earn users' trust, and incorporating this design can effectively optimize the overall experience of online customer service.

4.9 Overall Model

This section summarizes all hypothesis testing results of this study. Multiple linear regression analysis is adopted to explore the associations between the independent variables (Adhikari, 2022) and consumers' trust in AI chatbots in the online customer service context.

The regression analysis results of this study show that among the pre-set hypotheses, H1 (perceived usefulness), H3 (perceived interactivity), and H5 (anthropomorphism) are supported, as they have a significant correlation with consumer trust. Meanwhile, H2 (perceived ease of use) and H4 (perceived risk) are not supported, as their correlations lack statistical significance.

Table 4. 15: Summary Table of Hypothesis Testing Results

Hypothesis	**Relationship**	**Beta**	**Sig.**	**Result**
H1	PU→ DV	0.345	<0.001	Significant
H2	PEU→ DV	-0.010	0.910	Not Significant
H3	PI→ DV	0.194	0.020	Significant
H4	PR→ DV	0.104	0.119	Not Significant
H5	ANT→ DV	0.334	<0.001	Significant

The standardized beta coefficients from Table 4.15 of this study indicate that among the three core variables influencing consumers' trust in AI chatbots, the β values of perceived usefulness, anthropomorphism, and perceived interactivity are 0.345, 0.334, and 0.194 in that order. Chatbots with these traits are more likely to gain consumers' trust.

This study finds that the impacts of perceived ease of use and perceived risk on consumer trust do not reach statistical significance. The reason for this outcome is that the study's respondents are familiar with AI chatbots and online customer service systems, so related concerns are no longer a core consideration.

Multiple Linear Regression:

$$\text{Consumer Trust in AI Chatbots} =$$
$$-0.062 + 0.536\,(PU) - 0.015\,(PEU) + 0.284\,(PI) + 0.165\,(PR) + 0.457\,(ANT)$$

This study conducted regression analysis on AI chatbots used in online customer service scenarios. It found that four variables, including perceived usefulness, are positively correlated with consumer trust, while perceived ease of use is negatively correlated with consumer trust.

Among all tested variables, perceived usefulness and anthropomorphism have the strongest impact intensity.

4.10 Summary

This chapter presents all data analyses and core findings of the study on factors influencing consumers' trust in AI chatbots in the context of Malaysia's online customer service. This study uses SPSS Version 30.0 to test the associations between each of the study's independent variables and its core dependent variable.

This chapter focuses on a survey of AI chatbot usage behaviour. It first conducts an analysis of respondent profiles. The respondents of this study are mainly degree-seeking students, who frequently use this tool mostly in online customer service scenarios on social messaging applications and e-commerce platforms, and generally have sufficient interaction experience with the tool.

This study first completed a normality analysis; the skewness and kurtosis indicators all met the required standards, confirming the data approximately followed a normal distribution. A reliability test was then conducted, and all constructs had a Cronbach's Alpha value above 0.7, indicating good internal consistency.

The descriptive analysis of this survey shows that all respondents held positive perceptions of AI chatbots across the 6 observed variables. Among these perceptions, the mean score of consumer trust was the highest, and respondents accepted this service.

This study uses multiple linear regression to test the association between each core independent variable and consumer trust. The model is statistically significant, and it can explain 80.6% of the variance in consumer trust. Among the tested variables, perceived usefulness, perceived interactivity, and anthropomorphism produce significant positive effects, while perceived ease of use and perceived risk have no significant effects.

In conclusion, the research presented in this chapter confirms that the usefulness, interactivity, and human-like characteristics of online customer service AI chatbots can enhance consumer trust. The next chapter will discuss this study's implications, contributions, limitations, and future research directions.

CHAPTER 5

CONCLUSION AND RECOMMENDATION

5.1 Overview

In the previous chapter, we completed the full data analysis for the research on factors influencing consumer trust in AI chatbots for online customer service in Malaysia. We conducted four categories of statistical analysis relying on SPSS 30.0, and the results show that perceived usefulness, perceived interactivity, and anthropomorphism have significant effects on consumer trust, while perceived ease of use and perceived risk have no significant correlation with this outcome. This work lays an empirical foundation for the analysis in subsequent chapters.

This study focuses on the issue of consumer trust in AI chatbots in the online customer service scenario. As the opening introductory overview of the conclusion chapter, this chapter builds on the research objectives and hypotheses proposed in previous chapters. It will sequentially sort out the links between the study's core findings and existing literature and theories, elaborate on this study's theoretical and practical contributions, analyze its research limitations, and finally put forward directions for future research before concluding the entire chapter.

5.2 Research Findings and Discussion

This study aims to explore the factors influencing consumer trust in AI chatbots in the context of Malaysia's online customer service. Based on data collected from 240 respondents, it carries out its statistical analysis in Chapter 4.

First, this study conducted data screening and normality analysis. The results show that the collected samples are suitable for all subsequent analyses. The test indicators confirm that the distribution of respondents' answers is consistently normal, the samples are reliable, and they meet the requirements for the statistical tests used in this study.

The reliability analysis of this study shows that the Cronbach's alpha value of all variables are above the recommended value, which means that the measurement scales have satisfactory internal consistency and reliability. If Cronbach's alpha is greater than 0.70, it represents acceptable reliability in behavioral research (Hair et al., 2022). Likewise, Sarstedt et al. (2022) highlighted that the validity of the empirical results in studies on technology adoption depends on the availability of reliable measuring instruments.

Sample statistics from this research show that the majority of respondents are currently enrolled degree-seeking students aged 21 to 23, most of whom are from business or technology-related academic fields. In addition, all respondents have prior experience using AI chatbots for online customer service, and many respondents use such tools multiple times per week or even daily. Based on these observations, this study infers that this type of AI has become a routine component of consumers' online service experiences, a status that is particularly prominent among young groups proficient in digital technologies. This result aligns with the findings of recent studies which indicated that younger consumers tend to be more open to the use of AI-driven technologies due to their increased digital literacy and familiarity with online platforms (Herlina et al., 2025; Goel et al., 2022).

This study found through behavioral analysis that the core scenarios in which respondents use AI chat customer service are product inquiries and order tracking. Among access channels, social messaging applications including WhatsApp and Facebook Messenger account for the largest share, followed by e-commerce platforms such as Shopee and Lazada. Consumers' core needs are speed, convenience, and easy accessibility.

The descriptive analysis led by this study yields the core finding: For AI chatbots deployed in the online customer service scenario, this study measured five core variables, namely

usefulness, interactivity, ease of use, human-like features, and service trust. Respondents held relatively positive perceptions of most of the variables. Only the human-like feature dimension received moderate recognition, and the overall level of service trust also demonstrated a positive trend.

This paper conducts regression analysis and explores core influencing factors centered on the theme of consumers' trust in AI chatbots. It breaks down the core variable perceived usefulness into three dimensions aligned with consumption scenarios—solving user problems, providing useful information, and improving the efficiency of customer service interactions—and verifies that functional value is a key factor for building trust in AI technologies. This finding aligns with recent studies that have shown that perceived usefulness continues to be one of the most important factors to influence trust and acceptance in AI-powered customer service systems (Naveed, 2026). If the user feels that they can get something out of the conversations with the chatbot, he or she is more likely to have faith in it.

This study finds that perceived ease of use has no significant effect on consumers' trust in AI chatbots. Most respondents are familiar with digital technologies, so ease of use is no longer a core concern in their trust assessment. This was also observed by Goel et al. (2022) who reported that ease of use is not a significant factor when users have a significant amount of prior experience with digital technologies.

This study finds that the perceived interactivity of AI customer service chatbots significantly influences consumer trust. Users favor chatbots with timely responses, strong interactivity, and smooth communication, and an experience of high user engagement can reinforce user trust. The result aligns with existing research that indicates the interactive features of chatbots, like the video option, can boost customer satisfaction, engagement, and trust by fostering a more responsive customer support journey (Wang et al, 2026).

Building on the conclusions of the series of prior studies outlined earlier, this study finds that perceived risk has no significant effect on consumer trust. Although respondents are aware of concerns such as privacy and security, their daily habitual use of various digital services means these concerns do not strongly influence their trust in AI chatbots. This finding is consistent with recent research which indicated that the perceived risk would not affect trust formation among frequent users of digital platforms, as they might be more comfortable with the technology risks (Marza et al, 2019).

This study finds that anthropomorphism is strongly positively correlated with consumer trust. Survey respondents show greater willingness to trust AI chatbots that possess human-like interactive conversational characteristics. This was corroborated by Ktisti et al. (2026) who showed that when users are presented with human-like conversational cues, they feel more trusted, more competent and warmer. For enterprises, adding human-like communication functions to such chatbots can enhance users' trust in and acceptance of these tools (Luo et al, 2019).

To sum up, this study finds that AI chatbots with usefulness, interactivity, and a human-like communication experience are more likely to gain consumer trust. It also confirms that consumer acceptance of this technology in Malaysia is continuously rising, with the acceptance level among young, experienced digital users being particularly prominent.

5.3 Achievement of Research Objectives

This chapter relies on the completed data analysis and hypothesis testing results of this study and combines relevant findings from Chapter 4 and Section 5.2. It matches each corresponding research hypothesis one by one and evaluates whether all of this study's pre-set research goals have been met.

5.3.1 Objective 1: To Examine the Relationship Between Perceived Usefulness and Consumer Trust in AI Chatbots for Online Customer Service

The first research objective of this study is to examine the correlation between the perceived usefulness of AI chatbots in the online customer service context and consumer trust. When consumers recognize the benefits, efficiency, and experience optimization value of these chatbots, they will place greater trust in them. Clarifying this relationship can help enterprises develop AI customer service systems that align with consumer demands (Choung et al., 2023).

This study aligned its core objective with validating hypothesis H1, which posits that a significant association exists between perceived usefulness and consumer trust in AI chatbots. Per the multiple linear regression analysis detailed in Chapter 4, the two variables are significantly positively correlated, with a significance level of <0.001, leading to the determination that H1 is valid.

The empirical results of this study show that among all independent variables tested in this research, perceived usefulness is the strongest predictor of consumers' trust in AI chatbots (Prakash et al., 2023). If respondents believe that an AI chatbot can provide useful information, solve practical problems, and improve the efficiency of customer service interactions, they are more likely to develop trust in it (Ltifi, 2023). This demonstrates that the functional value of AI services is the core factor influencing trust.

The first pre-established goal of this study was successfully achieved. Empirical findings show that in the context of Malaysia's online customer service, perceived usefulness significantly affects consumers' trust in AI chatbots (Ding & Najaf, 2024).

5.3.2 Objective 2: To Examine the Relationship Between Perceived Ease of Use and Consumer Trust in AI Chatbots for Online Customer Service

The second core research objective of this study is to examine the association between the perceived ease of use of online customer service AI chatbots and consumer trust. Existing consensus shows that users generally trust technologies that are easy to operate and have low interaction costs, and these easy-to-use customer service chatbots will improve consumers' comfort and confidence when using them (Prakash et al., 2023).

Hypothesis H2, pre-formulated for this study, posits that perceived ease of use has a significant association with consumers' trust in AI chatbots. The multiple linear regression analysis conducted in Chapter 4 of this study returned a significance value of 0.910, leading to the final determination that H2 is not supported.

This study's empirical findings reveal that ease of use does not exert a strong impact on respondents' trust in AI chatbots, as most of the surveyed population are familiar with various types of digital technologies and AI systems, so ease of use is no longer a key consideration in their trust-related decision-making (Choung et al., 2023).

Although the correlations between the relevant variables in this study did not reach statistical significance, the core research objective was achieved: we completed the testing and identification of the association between the perceived ease of use of online customer service AI in Malaysia and consumer trust (Herlina et al., 2025).

5.3.3 Objective 3: To Examine the Relationship Between Perceived Interactivity and Consumer Trust in AI Chatbots for Online Customer Service

The third core objective of this study is to examine the association between consumers' perceived interactivity and consumer trust in the context of AI chatbots serving as online customer service. Chatbots that boast the traits of timely responses, attractiveness, and the ability to sustain smooth communication are more likely to earn consumers' trust (Prakash et al., 2023).

The third hypothesis proposed in this study, H3, was confirmed to hold through the multiple linear regression analysis conducted in Chapter 4. Perceived interactivity has a significant positive correlation with consumers' trust in AI chatbots, with a significance value of 0.020.

Observational findings from this study show that respondents have greater trust in AI chatbots that provide fast responses, effective communication, and interactive customer service experiences (Ltifi, 2023). These features can improve users' sense of engagement and comfort, ultimately strengthening their trust in the system.

The third pre-set objective of this study has been successfully achieved, as the perceived interactivity of Malaysian online customer service AI chatbots significantly impacts consumer trust (Ding & Najaf, 2024).

5.3.4 Objective 4: To Examine the Relationship Between Perceived Risk and Consumer Trust in AI Chatbots for Online Customer Service

The fourth core research objective of this study is to examine the association between consumers' perceived risk and their trust in AI chatbots in the online customer service scenario. When interacting with this type of AI customer service, consumers hold concerns over privacy, security, inaccurate information, and other related issues (Li et al., 2023). Clarifying this association can help institutions boost user confidence and reduce these relevant concerns.

The fourth pre-set research hypothesis of this study, H4, posits that there is a significant association between perceived risk and consumers' trust in AI chatbots. However, the multiple linear regression results presented in Chapter 4 show a significance value of 0.119, and this hypothesis is not supported.

This study's survey on user trust in AI chatbots shows that perceived risk did not have a strong impact on respondents' level of trust. One possible explanation is that most respondents are familiar with various types of digital services and online platforms and believe that the risks associated with using the tool are controllable (Ltifi, 2023). In addition, respondents prioritize the convenience and practicality of this tool over potential risks.

This study conducted a consumer survey targeting AI chatbots for online customer service in Malaysia. While the correlation between perceived risk and trust did not reach statistical significance, the study's core research objectives have been successfully achieved.

5.3.5 Objective 5: To Examine the Relationship Between Anthropomorphism and Consumer Trust in AI Chatbots for Online Customer Service

The fifth core objective of this study is to examine the association between anthropomorphism and consumers' level of trust in AI chatbots for online customer service. AI that has three anthropomorphic features—natural communication style, conversational interaction, and emotional expression—tends to receive positive responses from consumers and can also enhance the sense of comfort and connection felt during communication (Prakash et al., 2023). This study has clear research value.

This study conducted verification of the pre-established Hypothesis H5. Results from the multiple linear regression analysis detailed in Chapter 4 show that anthropomorphism has a significant positive correlation with consumer trust ($p < 0.001$), and Hypothesis H5 is therefore supported.

This study finds that respondents report higher levels of trust in AI chatbots that adopt anthropomorphic communication and possess human-like conversational characteristics (Ltifi, 2023). Incorporating anthropomorphic elements into the design of this type of chatbot can boost users' trust, engagement, and overall customer service experience.

This study has completed Research Objective 5: In the online customer service scenario in Malaysia, the anthropomorphism of AI chatbots significantly impacts consumer trust.

5.4 Knowledge Implications

The findings of this study enrich the existing body of research on consumer trust in AI chatbots within the online customer service context and fill the relevant research gap in the Malaysian setting. By testing the impacts of five core factors including perceived usefulness, perceived ease of use, perceived interactivity, perceived risk, and anthropomorphism, this study addresses the knowledge gap proposed in Chapter 1, remedies the limitations of prior research that mostly focused on topics such as technology adoption while rarely paying attention to trust, and supplements this field with new empirical evidence.

This study extends the classic Technology Acceptance Model (TAM) and trust theory and applies them to the scenario of AI chatbot services. Empirical analysis yields two core findings: First, the study confirms the main premise of TAM, the perceived usefulness of these AI technologies truly influences consumer confidence in the service, aligning with recent studies that have shown consumers are more likely to trust AI-powered products when they believe they can be used to better suit their needs (Goel, 2022). Second, anthropomorphism and perceived interactivity are contributing factors that have also been confirmed in recent studies, where having a human-like appearance and interactive responses are associated with greater user confidence in the credibility, social presence and trust of AI-driven chatbots (Kumar, 2026).

Within the research field of consumer trust in AI chatbots, this combination of technical and interpersonal interaction factors is consistent with the current state of AI research, revealing that consumer trust for AI is influenced by technical and social or relational attributes of the interaction (Brown et al, 2024), incorporating five core variables including perceived usefulness. It emphasizes the unique value of the anthropomorphism variable, highlights the core role of emotional and social factors, and broadens the academic community's understanding of the factors that influence trust in this field.

This study has prominent core methodological advantages: It adopts quantitative research and uses SPSS Version 30.0 to test associations between variables, with fully replicable operational procedures; the regression model achieves an 80.6% variance explanation rate for the consumer trust variable, indicating strong explanatory power of the model; and all selected respondents have actual experience interacting with AI chatbot services, endowing the sample with high reliability.

The results offer important suggestions for AI chatbot practitioners and developers, echoing the recent literature, in which trust is highlighted as a key factor in the successful implementation of AI chatbots in customer service settings (Dwevedi et al, 2023). This study confirms that consumers value practical, interactive, and anthropomorphic features far more than ease of use and perceived risk, and it fills gaps in existing understanding of the evolution of consumers' expectations of AI customer service technologies.

This study enriches the existing literature in the field of AI chatbots and consumer trust, fills the research gap related to the Malaysian context, and clarifies that functional anthropomorphic attributes are the core of trust building.

5.5 Managerial Implications

This study draws on the identified factors that influence consumers' trust in AI chatbots for online customer service and puts forward important management implications for all participants across the full industry chain in this field, including enterprises developing and rolling out such solutions, relevant organizations, and policymakers. These insights help all parties optimize the design and operation decisions of AI customer service systems.

This study's empirical findings show that perceived usefulness is the strongest factor influencing consumers' trust in AI chatbots. On this basis, enterprises should focus on their chatbots' functional performance and operational efficiency (Dwivedi et al, 2023 & Goel et al, 2022), refine core capabilities of providing accurate information, delivering fast responses, and solving difficult problems, and implement concrete actions including updating databases, connecting with customer service systems, and monitoring performance, to improve customer trust, satisfaction and long-term loyalty.

This study finds that the higher the interactivity of AI chatbots perceived by consumers, the stronger their trust in the brand. Businesses should prioritize developing high-interactivity AI chatbots (Ngo et al, 2025) and optimize their performance across three areas: real-time responses, coherent conversations, and adaptive replies, to reduce user frustration and build a smooth customer service experience.

The empirical findings of this study show that perceived risk did not significantly affect consumers' trust in AI customer service. However, privacy and security concerns related to these systems remain a core issue. Regulatory bodies across all countries must introduce relevant guidelines (Dwivedi et al, 2023), while enterprises must comply with these rules and disclose the entire process of customer data processing. These measures can boost public confidence in AI and promote the safe, widespread adoption of AI customer service.

The core finding of this study is that anthropomorphism has a significant impact on consumer trust. Based on this conclusion, this study puts forward practical anthropomorphism implementation strategies for enterprises to apply to their AI chatbots: adopting conversational language, personalized greetings, a friendly communication style, and natural interaction modes, as well as building the capability to identify customers' emotional preferences. Such human-like interactions can enhance users' sense of comfort and engagement (Araujo, 2021), ultimately strengthen consumer trust and optimize the overall customer experience.

Educational institutions and various types of organizations can apply the conclusions of this study to promote AI literacy and digital cognition among consumers (Ng et al, 2021). Leveraging the trait observed in this research that most respondents are already familiar with AI chatbots, these entities can carry out advanced education on responsible AI use and provide scenario-based exposure to related tools, helping users clarify the strengths and weaknesses of these AI systems.

This study provides actionable guidance for customer relationship management optimization, targeted at enterprises operating in the e-commerce and digital service sectors. The study recommends deploying AI chatbots, which can reduce operational costs, improve response efficiency, and deliver 24-hour customer support. However, AI chatbots may only act as a supplement to human customer service (Luo, 2019); complex problems that require empathy and human judgment must remain handled by human staff.

The findings of this study can provide valuable insights for enterprises and policymakers involved in the development and management of AI chatbots. In the online customer service scenario, focusing on practicality, interactivity, and human-like communication characteristics can enhance consumer trust and drive the widespread adoption of this technology.

5.6 Limitations of the Study

Although the present study has successfully completed all of its pre-set core research objectives, it still has several limitations that require public disclosure. These limitations may affect the generalizability of the study's conclusions and the validity of their interpretation, and we have adopted multiple measures to reduce these impacts.

The initial sample of this study consists of Malaysian university students aged 21 to 23 who are familiar with digital technologies and AI chatbot services. This sample structure means the study's conclusions cannot fully represent the entire population of Malaysia; in particular, it cannot adequately cover older consumers or groups with limited exposure to AI technology. The sample's high level of technology familiarity may be the reason why perceived ease of use, perceived risk, and consumer trust did not show a significant correlation in this study. To improve sample diversity, this study has included respondents from different colleges, programs, and educational backgrounds. Future similar studies should recruit respondents across a wider age range and with different levels of technical experience to enhance the generalizability of their conclusions.

This study adopts a quantitative research approach, using questionnaires as its core data collection tool. While this method can collect data from a large sample of respondents, its application in this study carries two types of limitations: first, response biases may arise because respondents answer questions based on their personal perceptions; second, some respondents did not fully understand the concept of AI chatbots. To address these issues, we adapted all survey items from established studies and drafted all items using concise, clear language.

This study included 5 independent variables, including perceived usefulness, perceived ease of use, perceived interactivity, perceived risk, and anthropomorphism. While this set of variables can explain most of the causes underlying consumers' trust in AI chatbots, it overlooks 5 additional potential influencing factors such as service quality and privacy concerns, and the existing framework cannot fully capture all dimensions that shape trust in this type of service.

This study has another limitation where it only adopts a cross-sectional research design, which collects all research data at a single time point. Consumers' perceptions and trust in AI chatbots will change alongside technological iteration and the accumulation of their usage experience, but cross-sectional data cannot capture these dynamic shifts. Follow-up longitudinal studies can be carried out to deeply explore the long-term development trajectory of this trust.

This study focuses exclusively on AI chatbots used in the online customer service scenario, and its conclusions cannot be extended to similar AI robots in other scenarios such as healthcare, education, banking, and public services, as the expectations, concerns, and factors that influence users' trust all differ across industries.

Although this study has certain limitations, it adopted standardized measures to ensure its reliability and validity and can provide effective insights for research on AI trust related to Malaysia's online customer service sector.

5.7 Recommendations for Future Study

Based on the limitations and core findings of this study, we propose future research directions for consumer trust in AI chatbots for online customer service.

All respondents in this study are individuals familiar with digital technologies and AI chatbots, which results in sample limitations. Future related studies may include respondents of different ages, occupations, and educational backgrounds, and specifically add older adults and people with limited technical experience to their samples, to conduct a more in-depth exploration of different groups' perceptions of and trust in these types of services.

This study proposes the second direction for future research, recommending that subsequent studies related to AI chatbot services adopt a mixed research method that integrates quantitative and qualitative approaches. Researchers will use questionnaires to collect large sample data and employ interviews or focus groups to deeply explore consumers' experiences, perceptions, and concerns, so as to clarify the root causes of their trust and related behaviours.

Future research on consumer trust in AI chatbots can, based on the five core variables this study focused on perceived usefulness, perceived ease of use, perceived interactivity, perceived risk, and anthropomorphism, incorporate additional potential variables including service quality and personalization, to enrich the dimensions of influencing factors in this field and improve relevant research models.

The authors of this paper propose that longitudinal study designs should be adopted for all future research on consumer trust in AI chatbots. User trust changes dynamically alongside users accumulated usage experience and technological iterations, so collecting data across multiple time periods can not only clarify the evolution of trust but also produce more precise conclusions regarding its long-term impacts.

This paper puts forward the fifth future research agenda in the field of AI chatbots. This study focuses solely on the online customer service scenario, and its conclusions cannot be directly applied across industries. This paper calls for follow-up research to expand the scope of investigation to multiple scenarios including healthcare, banking, education, tourism, and public services, compare the impacts of industry-related factors on consumer trust, and identify the unique challenges of real-world deployment.

The research recommendations proposed in this paper not only deepen the academic community's understanding of consumer trust in AI chatbots but also facilitate the improvement of AI customer service systems.

5.8 Overall Conclusions

This study was independently carried out by the author. It takes AI chatbots used in Malaysia's online customer service settings as its research subject. Based on the Technology Acceptance Model (TAM) and trust theory, the study tests five core variables. It analyzes data collected from 240 respondents using SPSS 30.0, meets all pre-set research goals, and offers practical references for consumer trust research in this field.

This study conducts an empirical test of the factors influencing consumer trust in AI chatbots. All five pre-determined influencing factors slated for testing passed the statistical significance test; among them, the core predictor's influence weight far exceeds that of all other factors. The conclusions drawn from this study can provide solid statistical support for AI-driven customer service systems to cultivate user trust and optimize user experience.

From two dimensions, the build-up of academic knowledge and the expansion of practical understanding, this study makes a unique contribution to research on consumer trust in AI chatbots in the Malaysian context. This study's conclusions can guide local businesses to build practical, highly interactive, human-like chatbots to increase customer trust. In addition, clarifying the core factors that shape trust is a key prerequisite for this technology to be successfully put into use and gain long-term adoption.

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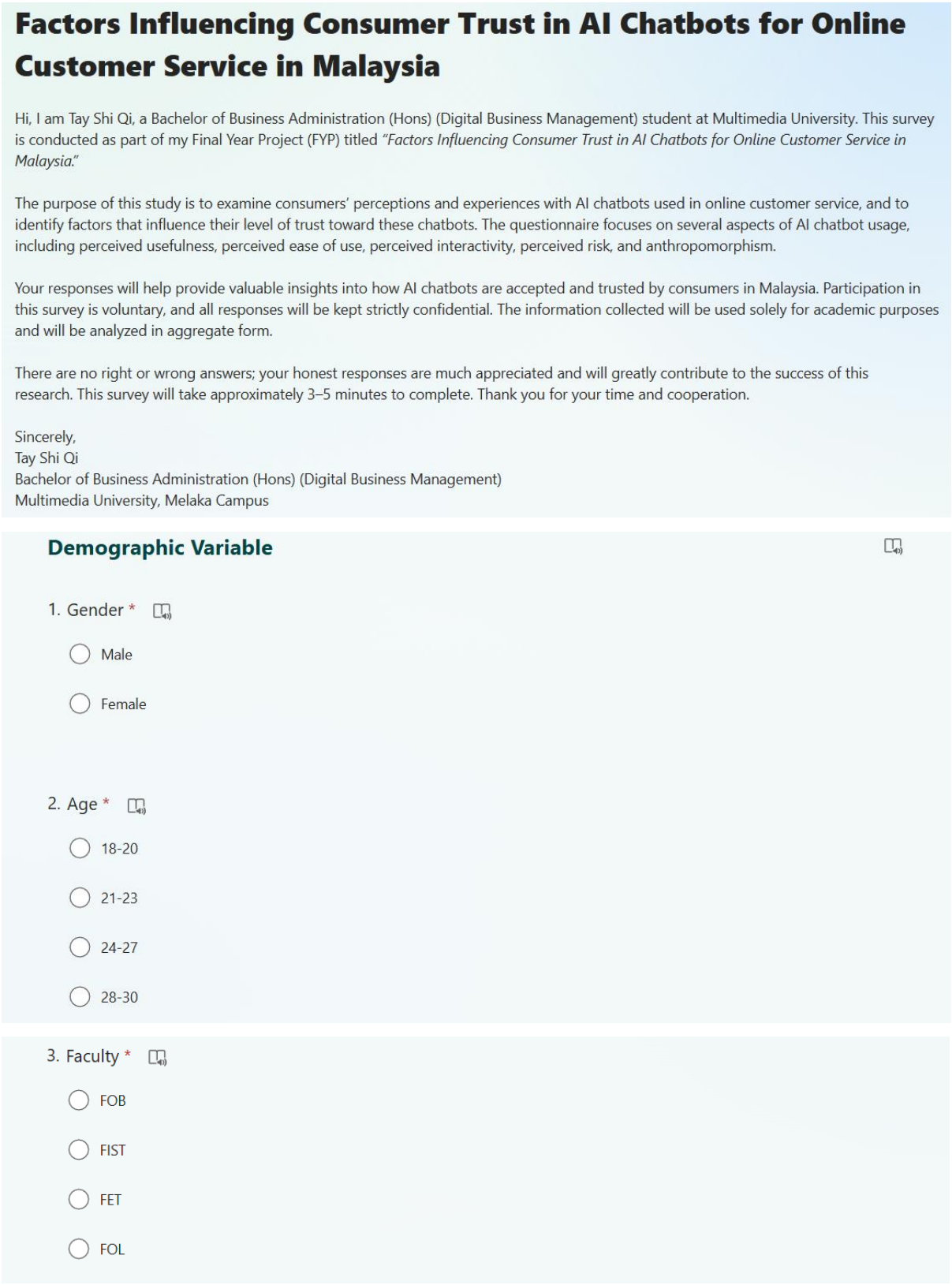
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Appendix

Form: https://forms.cloud.microsoft/r/wgcA8tx6Hd?origin=lprLink

Questionnaire Form



Factors Influencing Consumer Trust in AI Chatbots for Online Customer Service in Malaysia

Hi, I am Tay Shi Qi, a Bachelor of Business Administration (Hons) (Digital Business Management) student at Multimedia University. This survey is conducted as part of my Final Year Project (FYP) titled *"Factors Influencing Consumer Trust in AI Chatbots for Online Customer Service in Malaysia."*

The purpose of this study is to examine consumers' perceptions and experiences with AI chatbots used in online customer service, and to identify factors that influence their level of trust toward these chatbots. The questionnaire focuses on several aspects of AI chatbot usage, including perceived usefulness, perceived ease of use, perceived interactivity, perceived risk, and anthropomorphism.

Your responses will help provide valuable insights into how AI chatbots are accepted and trusted by consumers in Malaysia. Participation in this survey is voluntary, and all responses will be kept strictly confidential. The information collected will be used solely for academic purposes and will be analyzed in aggregate form.

There are no right or wrong answers; your honest responses are much appreciated and will greatly contribute to the success of this research. This survey will take approximately 3–5 minutes to complete. Thank you for your time and cooperation.

Sincerely,
Tay Shi Qi
Bachelor of Business Administration (Hons) (Digital Business Management)
Multimedia University, Melaka Campus

Demographic Variable

1. Gender *
   - ○ Male
   - ○ Female

2. Age *
   - ○ 18-20
   - ○ 21-23
   - ○ 24-27
   - ○ 28-30

3. Faculty *
   - ○ FOB
   - ○ FIST
   - ○ FET
   - ○ FOL

4. Programme *

- ◯ Banking and Finance
- ◯ Human Resource Management
- ◯ International Business
- ◯ Marketing Management
- ◯ Digital Business Management
- ◯ Data Communications and Networking
- ◯ Business Intelligence and Analytics
- ◯ Artificial Intelligence
- ◯ Security Technology
- ◯ Bioinformatics
- ◯ Electronic Engineering
- ◯ Mechanical Engineering
- ◯ Law
- ◯ Other

5. Education Level *

- ◯ Foundation
- ◯ Diploma
- ◯ Degree
- ◯ Postgraduate

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Behavioral Variable

6. Have you ever interacted with an AI chatbot for online customer service? *

- ◯ Yes
- ◯ No

7. How frequently do you use AI chatbots for online customer service? *

- ◯ Never
- ◯ Less than once a month
- ◯ 1–3 times a month
- ◯ 1–2 times a week
- ◯ Daily

8. For what purpose do you usually use AI chatbots? *

- ◯ Product inquiry
- ◯ Order tracking
- ◯ Complaints
- ◯ Payment or refund inquiries
- ◯ Technical support
- ◯ Other

9. Which platform do you usually use to interact with AI chatbots for online customer service? *

- ◯ E-commerce websites (e.g., Shopee, Lazada)
- ◯ Retail or service mobile applications (e.g., banking apps, airline apps)
- ◯ Social messaging applications (e.g., WhatsApp, Facebook Messenger, Telegram)
- ◯ Company official websites (live chat/chatbot feature)
- ◯ Email-based customer service with automated responses
- ◯ Other

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Dependent Variable

10. Consumer Trust in AI Chatbots *

	Strongly Disagree	Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Agree	Strongly Agree
I trust AI chatbots to provide reliable customer service.	○	○	○	○	○	○	○
I feel confident using AI chatbots for online customer service.	○	○	○	○	○	○	○
AI chatbots act in my best interest when handling my requests.	○	○	○	○	○	○	○
AI chatbots provide customer service in a trustworthy manner.	○	○	○	○	○	○	○

Back Next

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Independent Variables

11. Perceived Usefulness *

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
AI chatbots help me resolve customer service issues efficiently.	○	○	○	○	○
AI chatbots provide useful information for my online shopping needs.	○	○	○	○	○
AI chatbots save my time when seeking customer service support.	○	○	○	○	○
AI chatbots are useful for online customer service.	○	○	○	○	○

12. Perceived Ease of Use *

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
It is easy for me to interact with AI chatbots for customer service.	○	○	○	○	○
I find AI chatbots easy to understand and use.	○	○	○	○	○
Learning to use AI chatbots is easy for me.	○	○	○	○	○
AI chatbots are user-friendly.	○	○	○	○	○

13. Perceived Interactivity *

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
AI chatbots respond promptly to my questions.	○	○	○	○	○
AI chatbots allow two-way communication during service interactions.	○	○	○	○	○
AI chatbots provide relevant responses during interaction.	○	○	○	○	○
I feel involved when interacting with AI chatbots.	○	○	○	○	○
AI chatbots maintain a smooth and continuous conversation.	○	○	○	○	○

14. Perceived Risk *

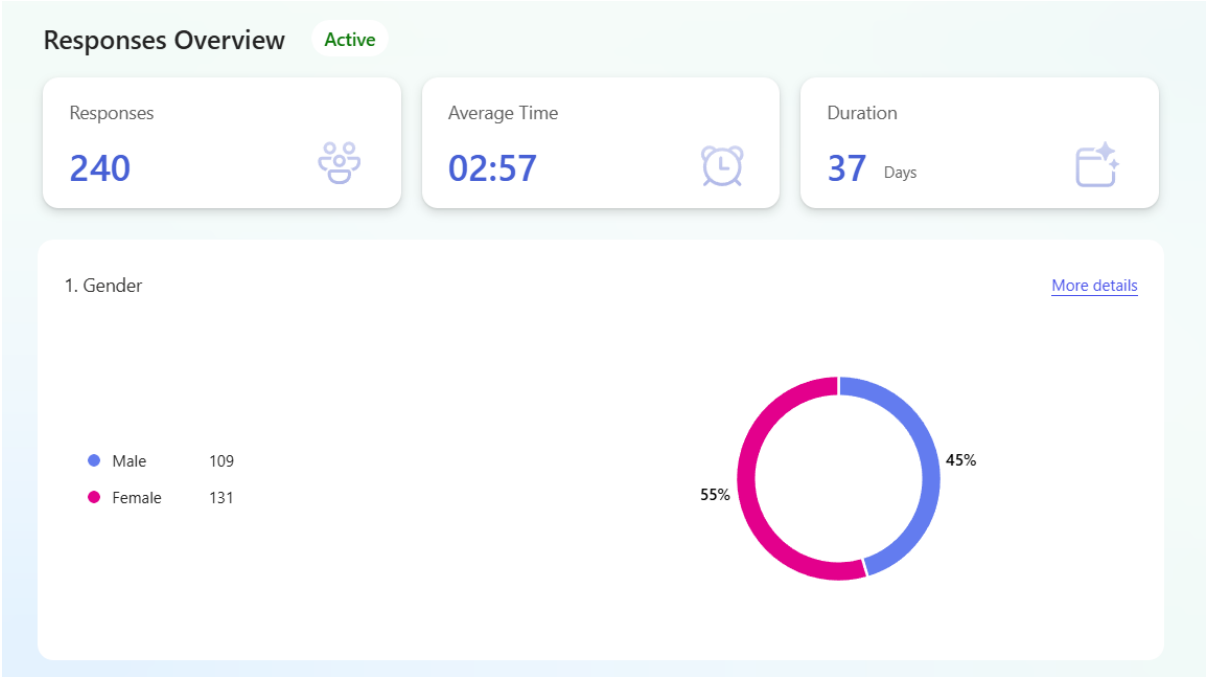
	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
I am concerned about my personal data security when using AI chatbots.	○	○	○	○	○
Using AI chatbots involves privacy risks.	○	○	○	○	○
I fear potential harm from my data being shared with third parties through AI chatbots.	○	○	○	○	○
I feel uncertain about the outcomes of using AI chatbots.	○	○	○	○	○
I worry about negative consequences when using AI chatbots.	○	○	○	○	○

15. Anthropomorphism (seeing a chatbot as human-like) *

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
AI chatbots communicate in a human-like manner.	○	○	○	○	○
AI chatbots appear friendly during interaction.	○	○	○	○	○
AI chatbots show understanding of my needs.	○	○	○	○	○
Interacting with AI chatbots feels similar to talking to a human.	○	○	○	○	○

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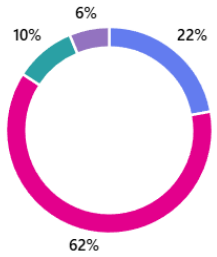
Data Results



2. Age

More details

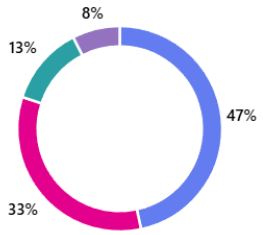
18-20	53
21-23	149
24-27	23
28-30	15



3. Faculty

More details

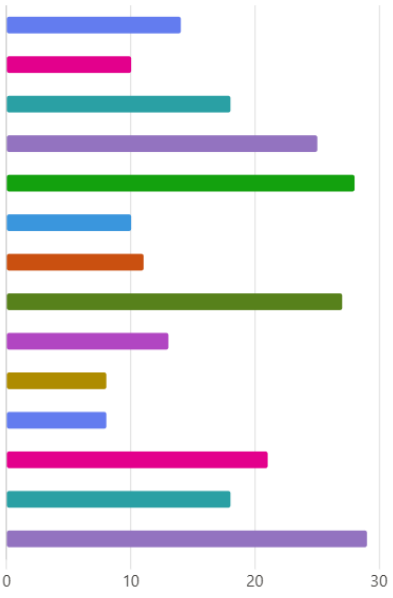
FOB	112
FIST	80
FET	30
FOL	18



4. Programme

More details

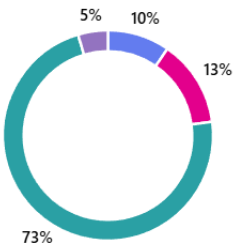
Banking and Finance	14
Human Resource Management	10
International Business	18
Marketing Management	25
Digital Business Management	28
Data Communications and Networking	10
Business Intelligence and Analytics	11
Artificial Intelligence	27
Security Technology	13
Bioinformatics	8
Electronic Engineering	8
Mechanical Engineering	21
Law	18
Other	29



5. Education Level

More details

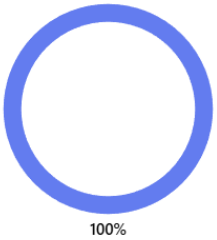
Foundation	23
Diploma	32
Degree	174
Postgraduate	11



6. Have you ever interacted with an AI chatbot for online customer service?

More details

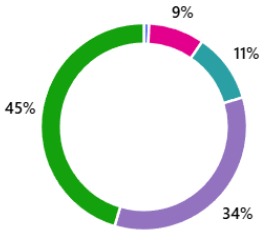
Yes	240
No	0



7. How frequently do you use AI chatbots for online customer service?

More details

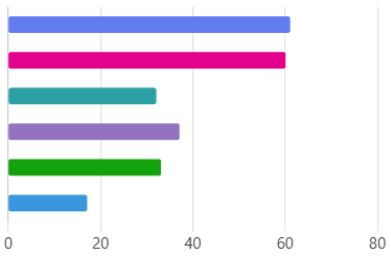
Never	2
Less than once a month	21
1–3 times a month	26
1–2 times a week	82
Daily	109



8. For what purpose do you usually use AI chatbots?

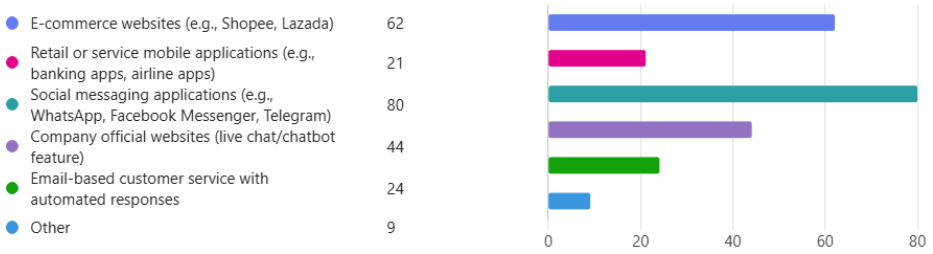
More details

Product inquiry	61
Order tracking	60
Complaints	32
Payment or refund inquiries	37
Technical support	33
Other	17



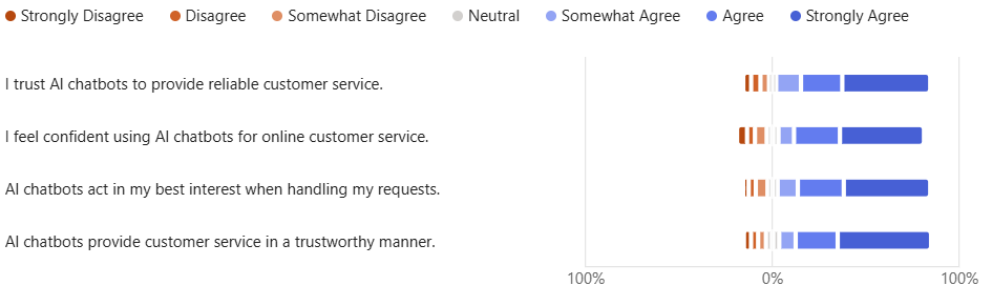
9. Which platform do you usually use to interact with AI chatbots for online customer service?

More details



10. Consumer Trust in AI Chatbots

More details



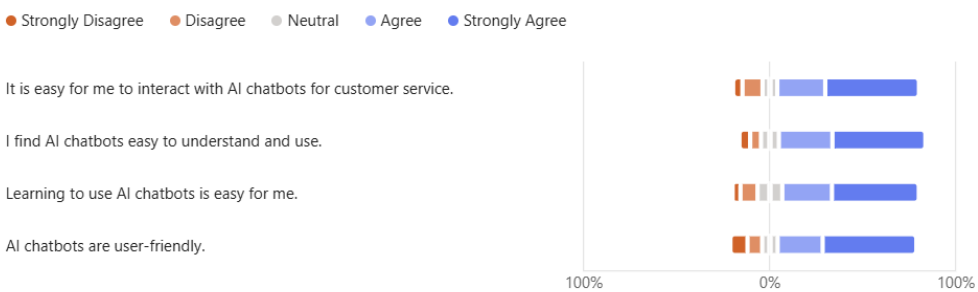
11. Perceived Usefulness

More details



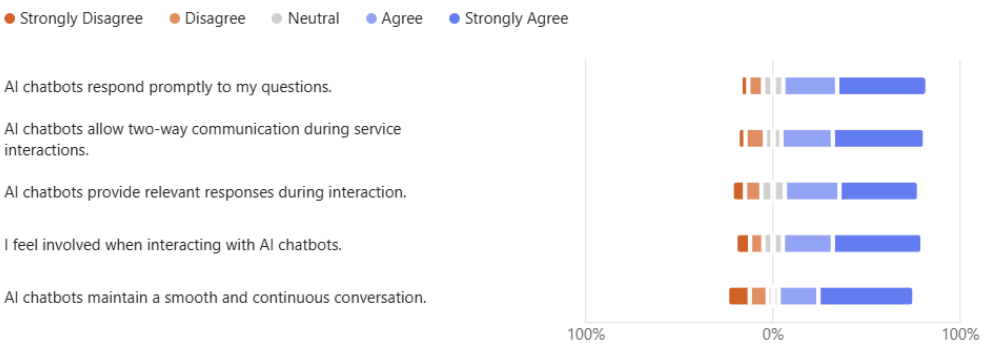
12. Perceived Ease of Use

More details



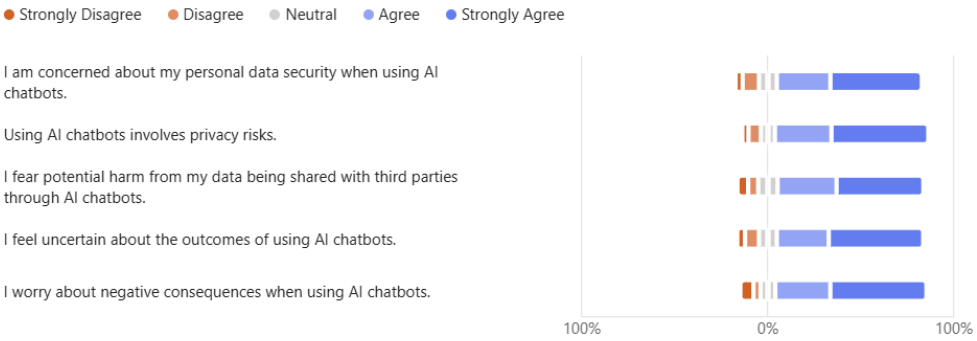
13. Perceived Interactivity

More details



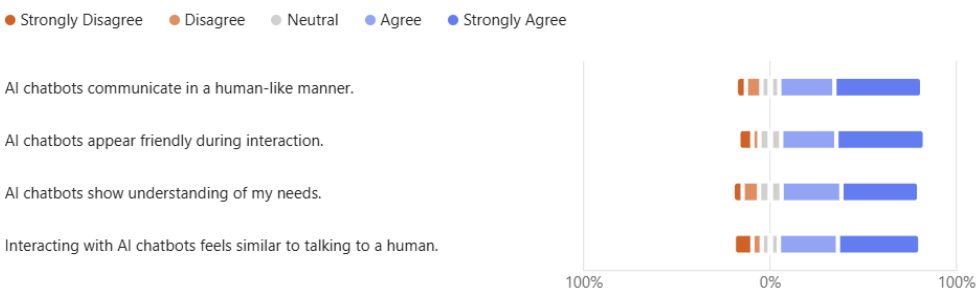
14. Perceived Risk

More details



15. Anthropomorphism (seeing a chatbot as human-like)

More details



SPSS Results

Frequency Table

Gender

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Male	109	45.4	45.4	45.4
	Female	131	54.6	54.6	100.0
	Total	240	100.0	100.0	

Age

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	18-20	53	22.1	22.1	22.1
	21-23	149	62.1	62.1	84.2
	24-27	23	9.6	9.6	93.8
	28-30	15	6.3	6.3	100.0
	Total	240	100.0	100.0	

Faculty

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	FOB	112	46.7	46.7	46.7
	FIST	80	33.3	33.3	80.0
	FET	30	12.5	12.5	92.5
	FOL	18	7.5	7.5	100.0
	Total	240	100.0	100.0	

Programme

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Banking and Finance	14	5.8	5.8	5.8
	Human Resource Management	10	4.2	4.2	10.0
	International Business	18	7.5	7.5	17.5
	Marketing Management	25	10.4	10.4	27.9
	Digital Business Management	28	11.7	11.7	39.6
	Data Communications and Networking	10	4.2	4.2	43.8
	Business Intellignece and Analytics	11	4.6	4.6	48.3
	Artificial Intelligence	27	11.3	11.3	59.6
	Security Technology	13	5.4	5.4	65.0
	Bioinformatics	8	3.3	3.3	68.3
	Electronics Engineering	8	3.3	3.3	71.7
	Mechanical Engineering	21	8.8	8.8	80.4
	Law	18	7.5	7.5	87.9
	Other	29	12.1	12.1	100.0
	Total	240	100.0	100.0	

Education_Level

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Foundation	23	9.6	9.6	9.6
	Diploma	32	13.3	13.3	22.9
	Degree	174	72.5	72.5	95.4
	Postgraduate	11	4.6	4.6	100.0
	Total	240	100.0	100.0	

BV1

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Yes	240	100.0	100.0	100.0

BV2

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Never	2	.8	.8	.8
	Less than once a month	21	8.8	8.8	9.6
	1-3 times a month	26	10.8	10.8	20.4
	1-2 times a week	82	34.2	34.2	54.6
	Daily	109	45.4	45.4	100.0
	Total	240	100.0	100.0	

BV3

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Product inquiry	61	25.4	25.4	25.4
	Order tracking	60	25.0	25.0	50.4
	Complaints	32	13.3	13.3	63.7
	Payment or refund inquiries	37	15.4	15.4	79.2
	Technical support	33	13.8	13.8	92.9
	Other	17	7.1	7.1	100.0
	Total	240	100.0	100.0	

BV4

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	E-commerce websites (e.g., Shopee, Lazada)	62	25.8	25.8	25.8
	Retail or service mobile applications (e.g., banking apps, airline apps)	21	8.8	8.8	34.6
	Social messaging applications (e.g., WhatsApp, Facebook Messenger, Telegram)	80	33.3	33.3	67.9
	Company official websites (live chat/chatbot feature)	44	18.3	18.3	86.3
	Email-based customer service with automated responses	24	10.0	10.0	96.3
	Other	9	3.8	3.8	100.0
	Total	240	100.0	100.0	

Descriptive Statistics for All Items

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
DV1	240	1	7	5.72	1.715
DV2	240	1	7	5.62	1.798
DV3	240	1	7	5.76	1.623
DV4	240	1	7	5.81	1.661
PU1	240	1	5	4.06	1.174
PU2	240	1	5	4.12	1.107
PU3	240	1	5	4.12	1.084
PU4	240	1	5	4.17	1.138

PEU1	240	1	5	4.06	1.212
PEU2	240	1	5	4.11	1.161
PEU3	240	1	5	4.02	1.160
PEU4	240	1	5	3.99	1.318
PI1	240	1	5	4.09	1.131
PI2	240	1	5	4.07	1.171
PI3	240	1	5	3.91	1.239
PI4	240	1	5	4.00	1.259
PI5	240	1	5	3.90	1.428
PR1	240	1	5	4.11	1.126
PR2	240	1	5	4.21	1.059
PR3	240	1	5	4.08	1.145
PR4	240	1	5	4.14	1.125
PR5	240	1	5	4.15	1.169
ANT1	240	1	5	4.05	1.179
ANT2	240	1	5	4.06	1.191
ANT3	240	1	5	3.96	1.174
ANT4	240	1	5	3.95	1.280
Valid N (listwise)	240				

Reliability Statistics

Reliability Statistics for Dependent Variable (DV)

Reliability Statistics

Cronbach's Alpha	N of Items
.898	4

Reliability Statistics for Perceived Usefulness (IV1)

Reliability Statistics

Cronbach's Alpha	N of Items
.873	4

Reliability Statistics for Perceived Ease of Use (IV2)

Reliability Statistics

Cronbach's Alpha	N of Items
.843	4

Reliability Statistics for Perceived Interactivity (IV3)

Reliability Statistics

Cronbach's Alpha	N of Items
.873	5

Reliability Statistics for Perceived Risk (IV4)

Reliability Statistics

Cronbach's Alpha	N of Items
.889	5

Reliability Statistics for Anthropomorphism (IV5)

Reliability Statistics

Cronbach's Alpha	N of Items
.922	4

Linear Regression

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Durbin-Watson
1	.898[a]	.806	.802	.66256	1.921

a. Predictors: (Constant), Anthropomorphism, PerceivedRisk, PerceivedUsefulness, PerceivedInteractivity, PerceivedEaseofUse

b. Dependent Variable: ConsumerTrust

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	426.952	5	85.390	194.516	<.001[b]
	Residual	102.723	234	.439		
	Total	529.675	239			

a. Dependent Variable: ConsumerTrust

b. Predictors: (Constant), Anthropomorphism, PerceivedRisk, PerceivedUsefulness, PerceivedInteractivity, PerceivedEaseofUse

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Collinearity Statistics	
		B	Std. Error	Beta			Tolerance	VIF
1	(Constant)	-.062	.200		-.309	.757		
	PerceivedUsefulness	.536	.100	.345	5.351	<.001	.200	5.007
	PerceivedEaseofUse	-.015	.133	-.010	-.113	.910	.103	9.695
	PerceivedInteractivity	.284	.121	.194	2.343	.020	.121	8.273
	PerceivedRisk	.165	.105	.104	1.564	.119	.189	5.295
	Anthropomorphism	.457	.071	.334	6.434	<.001	.308	3.243

a. Dependent Variable: ConsumerTrust

FACULTY OF BUSINESS

RESEARCH METHODS ASSIGNMENT (RMA) AND RESEARCH PROJECT (RP) MEETING RECORD FORM

Student Name:	**TAY SHI QI**	Student ID Number:	**251UB2511X**
Meeting Number:	**1 & 2**	Date of Meeting:	**2 April 2026 & 7 April 2026**
Supervisor:	**Mr. Tan Seng Huat**	Co-Supervisor (If any):	**-**
Project Title:	**Factors Influencing Consumer Trust in AI Chatbots for Online Customer Service in Malaysia**		

*Note: Student is required to attach at least **5** "RMA and RP Meeting Record Form"s in the RMA/RP final report.*

Issues raised by student:

Request for checking questionnaire form.

Solutions suggested by supervisor:

DV- Use 7 Likert scale. Add "neutral"
IV- Use 5 Likert scale.

Actions taken by student (after the meeting):

Proceed with data collection

Student: Tay Shi Qi	Supervisor: Tan Seng Huat	Supervisor's Remarks (if any):
Signature: *shizi*	Stamp and Signature: TAN SENG HUAT Lecturer Faculty Of Business (FOB) Multimedia University Jalan Ayer Keroh Lama 75450 Melaka	
Date: 7 April 2026	Date: 7 April 2026	

FACULTY OF BUSINESS

RESEARCH METHODS ASSIGNMENT (RMA) AND RESEARCH PROJECT (RP) MEETING RECORD FORM

Student Name:	**TAY SHI QI**	Student ID Number:	**251UB2511X**
Meeting Number:	**3 & 4**	Date of Meeting:	**24 April 2026 & 6 May 2026**
Supervisor:	**Mr. Tan Seng Huat**	Co-Supervisor (If any):	**-**
Project Title:	**Factors Influencing Consumer Trust in AI Chatbots for Online Customer Service in Malaysia**		

*Note: Student is required to attach at least **5** "RMA and RP Meeting Record Form"s in the RMA/RP final report.*

Issues raised by student:

Data collection is done, and the responses are being reviewed in Microsoft Forms. SPSS data entry has not started yet.

Solutions suggested by supervisor:

Proceed for SPSS data estimation.
1. Frequency analysis
2. Reliability test
3. Linear regression
4. Normality test
5. Pearson (optional)

Actions taken by student (after the meeting):

Key in SPSS and share the data to supervisor.

Student: Tay Shi Qi	Supervisor: Tan Seng Huat	Supervisor's Remarks (if any):
Signature: *shizi*	Stamp and Signature: TAN SENG HUAT Lecturer Faculty Of Business (FOB) Multimedia University Jalan Ayer Keroh Lama 75450 Melaka	
Date: 6 May 2026	Date: 6 May 2026	

FACULTY OF BUSINESS

RESEARCH METHODS ASSIGNMENT (RMA) AND RESEARCH PROJECT (RP) MEETING RECORD FORM

Student Name:	**TAY SHI QI**	Student ID Number:	**251UB2511X**
Meeting Number:	**5 & 6**	Date of Meeting:	**7 May 2026 & 11 May 2026**
Supervisor:	**Mr. Tan Seng Huat**	Co-Supervisor (If any):	**-**
Project Title:	**Factors Influencing Consumer Trust in AI Chatbots for Online Customer Service in Malaysia**		

*Note: Student is required to attach at least **5** "RMA and RP Meeting Record Form"s in the RMA/RP final report.*

Issues raised by student:

Request for reviewing and checking the SPSS results.
R square=0.948
All VIF>10

Solutions suggested by supervisor:

Check value for R square and VIF due to the high value of the results. [some mistakes during the estimation]

Check the mean from descriptive statistics.
DV 1-7 Likert scale: mean range around 5
IV1 to IV5 1-5 Likert scale: mean range around 3-4
Results are positive.

Actions taken by student (after the meeting):

Proceed with Chapter 4 and 5.

Student: Tay Shi Qi	Supervisor: Tan Seng Huat	Supervisor's Remarks (if any):
Signature: *shizi*	Stamp and Signature: TAN SENG HUAT Lecturer Faculty Of Business (FOB) Multimedia University Jalan Ayer Keroh Lama 75450 Melaka	
Date: 11 May 2026	Date: 11 May 2026	

FACULTY OF BUSINESS

RESEARCH METHODS ASSIGNMENT (RMA) AND RESEARCH PROJECT (RP) MEETING RECORD FORM

Student Name:	**TAY SHI QI**	Student ID Number:	**251UB2511X**
Meeting Number:	**7 & 8**	Date of Meeting:	**6 June 2026 & 11 June 2026**
Supervisor:	**Mr. Tan Seng Huat**	Co-Supervisor (If any):	**-**
Project Title:	**Factors Influencing Consumer Trust in AI Chatbots for Online Customer Service in Malaysia**		

*Note: Student is required to attach at least **5** "RMA and RP Meeting Record Form"s in the RMA/RP final report.*

Issues raised by student:

Done Chapter 4 and 5, request for checking.

Solutions suggested by supervisor:

1.3 Research Problem – Split the paragraph
Delete secondary data from 3.6.2

Chapter 4 table need to redo
Chapter 5 Objectives need to add citation

Actions taken by student (after the meeting):

Follow what the supervisor suggested.

Student: Tay Shi Qi	Supervisor: Tan Seng Huat	Supervisor's Remarks (if any):
Signature: *shiqi*	Stamp and Signature: TAN SENG HUAT Lecturer Faculty Of Business (FOB) Multimedia University Jalan Ayer Keroh Lama 75450 Melaka	
Date: 11 June 2026	Date: 11 June 2026	

FACULTY OF BUSINESS

RESEARCH METHODS ASSIGNMENT (RMA) AND RESEARCH PROJECT (RP) MEETING RECORD FORM

Student Name:	**TAY SHI QI**	Student ID Number:	**251UB2511X**
Meeting Number:	**9**	Date of Meeting:	**23 Jun 2026**
Supervisor:	**Mr. Tan Seng Huat**	Co-Supervisor (If any):	**-**
Project Title:	**Factors Influencing Consumer Trust in AI Chatbots for Online Customer Service in Malaysia**		

*Note: Student is required to attach at least **5** "RMA and RP Meeting Record Form"s in the RMA/RP final report.*

Issues raised by student:

Request for re-checking FYP

Solutions suggested by supervisor:

Add more citation

Actions taken by student (after the meeting):

Add citation

Student: Tay Shi Qi	Supervisor: Tan Seng Huat	Supervisor's Remarks (if any):
Signature: *shiqi*	Stamp and Signature: TAN SENG HUAT Lecturer Faculty of Business (FOB) Multimedia University Jalan Ayer Keroh Lama 75450 Melaka	
Date: 23 June 2026	Date: 23 June 2026	

FYP2

ORIGINALITY REPORT

14%	9%	11%	%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	Jinjie Li, Lianren Wu, Jiayin Qi, Yuxin Zhang, Zhiyan Wu, Shuaibo Hu. "Determinants Affecting Consumer Trust in Communication With AI Chatbots", Journal of Organizational and End User Computing, 2023 Publication	2%
2	www.mdpi.com Internet Source	1%
3	Ashish Viswanath Prakash, Arun Joshi, Shubhi Nim, Saini Das. "Determinants and consequences of trust in AI-based customer service chatbots", The Service Industries Journal, 2023 Publication	1%
4	Al Juhani, Bayan. "The Impact of AI Chatbot on Customer Loyalty Mediating by Customer Satisfaction", Princess Sumaya University for Technology (Jordan) Publication	<1%
5	umkeprints.umk.edu.my Internet Source	<1%
6	"Big Data and Social Computing", Springer Science and Business Media LLC, 2024 Publication	<1%
7	www.airs.icbt.lk Internet Source	<1%